

Investigating Aneuploidy Events Associated with Immunosuppressive Phenotype in Ovarian Cancer

Authors: LingShan Hung¹, Haonan Lu^{2*}

Affiliations:

¹ LingShan Hung, Imperial Centre for Translational and Experimental Medicine, Department of Surgery and Cancer, Imperial College London, Hammersmith Hospital, London, United Kingdom.

² Haonan Lu, Imperial Centre for Translational and Experimental Medicine, Department of Surgery and Cancer, Imperial College London, Hammersmith Hospital, London, United Kingdom.

² Haonan Lu, Department of Obstetrics & Gynaecology, The University of Hong Kong, Hong Kong, China.

Student CID number: 01877777

Statement of Originality: I hereby certify that this submission is my own work, except where authorized collaboration is indicated, and I understand that any form of Academic Misconduct is strictly forbidden and may result in disciplinary action.

Abstract

Background: Aneuploidies, led by chromosome arm-level copy number alterations, are commonly observed in ovarian cancers, yet their roles in dysregulation of anti-tumour immunity remain poorly defined. We therefore hypothesized that aneuploidy play a role in cancer immune microenvironment. This study aims to identify immune associated arm-level copy number alterations and uncover its underlying molecular mechanism.

Methods: Utilizing bulk RNA sequencing data from The Cancer Genome Atlas and Hammersmith Hospital, we applied computational tools including FACETS, GISTIC2, and BISCUT to decode cancer copy number change and for in-depth genomic analysis. Pattern of arm-level copy number alterations were associated with immune markers, cell density, patient survival, and treatment response to identify potential immune-related arm-level copy number alterations. Single cell sequencing (scRNA-Seq) data from 42 HGSOc patients were utilised to explore the underlying molecular mechanism.

Results: chromosome 2q amplification in ovarian cancer was consistently associated with an immunosuppressive phenotype, reduced immune cell infiltration, and worse prognosis. Single-cell analyses highlighted a suppression of interferon signalling and altered cellular communication, suggesting 2q amplification could play a significant role in immune escape mechanisms in ovarian cancer.

Conclusion: Our findings identifies that 2q amplification is associated with immune evasion in ovarian cancer, providing a novel perspective on the genomic basis of cancer immunology. The study advances our understanding of aneuploidy in tumorigenesis and paves the way for future work to explore therapeutic targets.

Background

Copy Number Alterations in Cancer

Cancer arises from the accumulation of genetic mutations providing selective proliferative advantage upon cells, leading to unchecked proliferation and tumorigenesis. The size of these mutations ranges from point mutations to copy number alterations (CNA). CNA is a broad category embracing chromosome-level, arm-level (i.e., amplification or deletion of chromosome arms), and focal-level change (i.e., CNA of chromosomal segments smaller than an entire chromosome arm, usually encompassing panels of 10-30 genes) (1). CNA could also be categorised as either germline or arising somatically, with former usually denoted as copy number variation (CNV) while latter usually called either just CNA or somatic CNA (SCNA). In recognition of the diverse types of copy number alteration events and accompanying terminological variations, this study adopts the term aneuploidy to describe arm-level SCNA events.

Although present in around 90% of solid malignancies, the exact mechanisms by which aneuploidy contribute to oncogenesis remain incompletely understood (2, 3). One key reason is that in contrast to the well-recognised role of canonical oncogenes and tumour suppressor genes (e.g., TP53 or HER2), the oncogenic potential of aneuploidy is context dependent. Aneuploidy could suppress or promote tumorigenesis depending on cancer stage, histology, tumour microenvironment, and immune interactions (1, 4).

The prevalence of these aneuploidy also differs between cancer types, with certain malignancies such as thyroid carcinoma presenting aneuploidy in merely a quarter of the sampled cellular population, whereas examples like ovarian and lung cancers demonstrate over 80% cellular involvement in copy number alterations (3). Even within ovarian cancer, elevated aneuploidy score is predominantly associated with advanced-stage disease, increased age, and high-grade serous carcinoma subtype (5).

Several theories currently exist in explaining how aneuploidy promotes tumorigenesis. Aneuploidy induces genomic instabilities, such as chromosome instability (CIN) and microsatellite instability (MIN), often by replication stress or chromothripsis (6). These alterations enable rapid cancer evolution, creating more resilient subpopulations (6-8). Aneuploidy disrupt gene expression on a large scale, leading to proteotoxic stress, which involves protein folding issues and increased autophagy (6, 9). Cells with aneuploidy often exhibits a special phenotype with downregulated nucleic acid metabolism and upregulated autophagy and inflammation, reshaping the tumor microenvironment and contributing to tumorigenesis (6).

Lung and Ovarian Cancer

Ovarian cancer has a global age-standardized incidence rate of 6.6 and a mortality rate of 4.2 per 100,000 individuals (10). Predominantly characterized by high-grade serous carcinoma, this subtype accounts for 60% of epithelial ovarian cancer cases. Lung cancer remains the leading cause of cancer mortality worldwide, with a five-year survival rate of 15% for non-small cell lung cancer (11) and a marginally better prognosis of 25-30% for small cell variants (12).

Studying aneuploidy in ovarian and lung cancer is essential due the high prevalence of copy number alterations in both malignancies (5, 13-15). In ovarian cancer, Aneuploidy can provide insights into the aggressive nature and treatment resistance of the disease (5). In lung cancer, understanding aneuploidy is crucial for the development of targeted therapies, particularly in non-small cell lung cancer (NSCLC) types where these genetic changes can influence tumor behaviour and therapeutic outcomes (16). Aneuploidy can also affect how cancer cells interact with the immune system, for instance via overexpression of immune checkpoint molecules or loss of tumour antigen, potentially making them more adept at evading immune surveillance. Thus, researching aneuploidy offers insights into mechanisms of immune evasion and potential therapeutic targets to enhance immune response in these cancers.

Materials and Methods

Bulk RNA sequencing data

The transcriptomic landscapes of ovarian and lung cancer were dissected through a comprehensive analysis consisting of both discovery and validation cohorts. The discovery dataset encompassed lung cancers (both lung adenocarcinoma (LUAD) and squamous lung carcinoma (LUSC)) and ovarian malignancies, sourced from The Cancer Genome Atlas (TCGA). We downloaded the segmented copy number variation data, RNA sequencing data standardized via the PANCAN method, alongside detailed patient phenotypic and survival metrics, retrieved from the UCSC Xena platform (<https://tcga-xena-hub>) and cBioPortal database (<https://www.cbioportal.org>). Copy number

profiling was conducted using the Affymetrix Genome-Wide Human SNP Array 6.0, referencing the hg19 genome assembly (3). Patient selection was meticulously curated through a comparison across RNA sequencing datasets, copy number segmentation files, and clinical matrices, culminating in the inclusion of 954 lung cancer and 574 ovarian cancer patients for the final analysis. The lung cancer subset was characterized by 386 female and 566 male patients, ages spanning from 38 to 90 years, subdivided into 485 LUAD and 469 LUSC cases. The ovarian cancer cohort encapsulated 670 patients, with age distributions between 26 and 89 years. The validation cohort consists in-house collected datasets from NSCLC and high-grade serous ovarian cancer (HGSOC) at London's Hammersmith Hospital with publicly available data from NSCLC patients catalogued in the MSKCC IMPACT sequence database, who had undergone anti-PD-(K)1 treatment (17). The Hammersmith dataset utilized the SureSelect Human All Exon V6 r2 kit for library preparation, with sequencing conducted on the BGI DNBSEQ platform, using the hg38 genome assembly as the reference. Additionally, the MSKCC IMPACT sequence dataset—incorporating both IMPACT and WES sequencing technologies—was procured, including copy number segmentation and a comprehensive clinical information dataframe from cBioPortal detailing survival, treatment response, and immune characteristics. The Hammersmith collection comprised 269 original samples from 174 patients, delineated into 119 HGSOC and 55 NSCLC cases, with the HGSOC patients' ages ranging from 30 to 91, and NSCLC patients from 38 to 80, including 30 males and 25 females. The MSKCC cohort included 240 patients, with a gender distribution of 118 males and 122 females, aged between 22 and 92 years (Supplementary Figure 1A-C).

Hammersmith Dataset Sequencing Metrics

In this study, we built a customized hg38.fasta reference by selecting only the chromosomal and genomic sequences found in the BAM files, as listed in the Supplementary Method. This reference was created using data from the UCSC GoldenPath GRCh38 directory (Supplementary Method). We also prepared necessary index files with Samtools (v1.14) and dictionary files with Picard (v2.6.0) to accompany the reference. Target enrichment design files were obtained from Agilent Technologies, leading to the creation of target interval lists. This was done by applying the Picard BedToIntervalList to the hg38.dict and an edited SureSelect_Covered.bed file that had its header removed. From these target lists, we generated bait interval lists using Picard BaitDesigner.

For quality control, we used Picard HsMetrics to gather whole-exome sequencing (WES) statistics. The initial quality checks aimed to ensure mean and median target coverage of over 50x, and GC bias was evaluated using Picard CollectGcBiasMetrics, shown in Supplementary Figures 1D-E.

FACETS

The facetsSuite (v2.0.8) is an R package that simplifies the use of FACETS (v0.6.2), a tool for calling allele-specific copy numbers from paired tumor-normal DNA sequencing data produced by next-generation sequencing technologies, such as WES (18). FACETS operates as an independent pipeline, beginning with sequencing BAM files. It counts

allele-specific reads for both homozygous and heterozygous SNPs to ensure no detail is lost (19).

In our study, we worked with both matched and unmatched BAM files. For those that were matched, FACETS used its own SNP-pileup feature to create read count files from dbSNP build 151, following specific parameters outlined in the developer's guide “-g -q15 -Q20 -P100 -r25,0”. In the case of unmatched BAM files, we selected a random BAM file and processed it to remove germline CNVs, serving as a control for the SNP-pileup process.

Copy number analysis was then performed using the facetsSuite wrapper script, which was set with purity critical values (cval) of 100 for the initial run and 50 for a subsequent run, using the hg38 reference genome. This script executes FACETS in a two-stage process: the first stage estimates the overall segmentation profile, sample purity, and ploidy, while the second, more sensitive stage, aims to uncover more localized changes. FACETS provides estimates of tumor purity and ploidy, as seen in Supplementary Figure 1F. It also visualizes chromosome copy number states, as shown in Supplementary Figure 1G. The output from the second stage of analysis, specifically the adjusted dipLogR segmentation files for each paired sample, was combined to create a comprehensive segmentation file ready for analysis with GISTIC2.

GISTIC2.0

We employed GenePattern GISTIC2.0 (v6.15.30) analysis to identify recurrent focal and arm-level somatic copy number alterations (aneuploidy). We supplemented the segmentation data with a reference gene file in .mat format acquired from the Broad Institute's resource disk (20) and applied a specific parameter set derived from literature recommendations (21). The configuration included setting the amplification and deletion thresholds, as well as the q-value threshold, to 0.25. We also specified a joint segment size of 4, a cap value of 1.5, a confidence level of 95%, and a focal length cutoff of 0.50. The analysis incorporated advanced features such as arm peeling, broad analysis, and gene GISTIC, utilizing the extreme gene collapse method to streamline the identification of significant aneuploidy.

Immune markers

We curated a panel of sentinel immune response markers, encompassing cytotoxic genes such as *CD274* (PD-L1), *GNLY*, *GZMB*, and *PRF1* (22). To quantify the expression of MHC I and MHC II, we calculated the mean expression levels of their constitutive genes — *HLA-A*, *HLA-B*, and *HLA-C* for MHC I; and *HLA-DRB1*, *HLA-DQB1*, *HLA-DPB1*, *HLA-DRB5*, *HLA-DOB*, and *HLA-DQB2* for MHC II, respectively. Tertiary Lymphoid Structures (TLS) were estimated through the average expression of a cluster of specific genes including *CD79B*, *CD1D*, *CCR6*, *LAT*, *SKAP1*, *CETP*, *EIF1AY*, *RBP5*, *PTGDS*, *CCL19*, *CCL21*, *CXCL13*, *CCR7*, *CXCR5*, *SELL*, and *LAMP3* (23). Immunohistochemistry (IHC)-derived cell densities for T cells, B cells, and monocytes were estimated by MCPcounter (v1.2.0) algorithm from gene expression (24). Lastly, calculated tumor mutation burden (TMB) scores were retrieved from the clinical matrix available on the cBioPortal for TCGA.

Survival Analysis

Survival outcomes were evaluated using Cox proportional hazards regression, with continuous copy number variation

serving as the predictor of patient survival duration. The survival metric was derived via 'Surv' from the survival (v3.5-8), which integrates both the survival time (measured in days) and the survival event status (0 denoting survival, 1 denoting death). Kaplan-Meier survival curves were generated employing 'ggsurvplot' from the survminer (v0.4.9), facilitating visualization of survival distributions. Significance testing for differences between survival curves was executed utilizing the log-rank test, implemented within the package framework, with weights parameterized according to 'survdiff' (weights set to 1). This methodology allowed for robust statistical comparisons of survival endpoints across genomic profiles.

Single Cell Sequencing Data Pre-processing

Single-cell RNA sequencing data (scRNA-Seq) for ovarian cancer was downloaded from Vázquez-García (25) in the format of Seurat object files stored as RDS on Synapse repository (<https://datasets.cellxgene.cziscience.com>). This dataset comprises 929,686 cells from 156 specimens procured from 41 patients.

To ensure compatibility with the latest analytical frameworks, we extracted the raw count matrices and associated metadata from the original Seurat objects, reconstructing them within Seurat (v5.0.0). During pre-processing, we refined the dataset by excluding cells expressing fewer than 200 genes and genes present in fewer than 10 cells. Further quality control was executed via 'Add_Cell_QC_Metrics' from scCustomize (v2.1.2) package, eliminating cells with mitochondrial gene expression exceeding 15% and a complexity index ($\log_{10}[\text{GenesPerUMI}]$) falling below 0.75 (Supplementary Figure 2A-D).

Normalization of expression matrices was conducted through log normalization using Seurat, followed by scaling to mitigate the influence of cell-cycle genes, quantified alongside other quality metrics such as mitochondrial gene expression and unique molecular identifier (UMI) counts per cell. Subsequently, the 3000 most variable genes were identified to facilitate principal component analysis (PCA) on the Seurat object. The decision of selecting top 15 principal components for further analysis was guided by an ElbowPlot, with chosen components input into 'FindClusters' to identify cellular clusters (Supplementary Figure 2E). Cluster resolution was fine-tuned using the clustree (v0.5.0), with an initial resolution set at 0.5 to delineate major cell populations. These clusters were re-annotated using previously extracted metadata. Beyond the canonical Seurat analysis pipeline, we integrated a harmonization step to address identifiable batch effects, as evidenced by dimensionality plots coloured by sample ID (Supplementary Figure F). This harmonization process, implemented via the Harmony (v1.2.0) algorithm, corrected for variables such as tumor anatomical subsite, sample ID, and CD45 expression status—all derived from the metadata. The efficacy of this adjustment was visually confirmed by the absence of batch effects in subsequent dimensional analyses (Supplementary Figure G), illustrating the refined and integrative approach adopted in our analysis. A separate lung cancer single cell sequencing dataset was also procured for analysis (26). However, due to time constraints, the investigation was paused following quality control and the preliminary reconstruction of the Seurat

object, pending further exploration (Supplementary Figure 3A-H).

Arm-level Copy Number Estimation for Single Cell Sequencing Data

Arm-level somatic copy number alterations (Aneuploidy) were identified through two distinct methodologies. In instances where whole-genome sequencing (WGS) data were accessible (25), the GISTIC2 algorithm was employed to analyse copy number segmentation files generated from WGS, as illustrated in Supplementary Figure 2H. Conversely, for cases limited to single-cell sequencing data (26), we leveraged InferCNV (1.19.1) to extract aneuploidy from the patients' single-cell transcriptomic landscapes. Here, non-malignant cells, including endothelial cells, fibroblasts, and immune cells, served as reference points to delineate Aneuploidy within neoplastic cells. To address dynamic noise inherent in these analyses, a sigmoidal filtering approach was adopted, preferentially attenuating signals proximal to the mean intensity over those at the periphery. Specifically, the logistic function's midpoint was set at a standard deviation amplifier of 3, with a recommended cutoff threshold of 0.1 tailored for data generated via 10x Genomics platforms.

Our analytical framework incorporated a six-state CNV model to accurately determine levels of CNVs. This strategy facilitated the organization of samples into clusters based on the 'patient_celltype' categorization, diverging from the conventional 'k_obs_groups' classification. Regions identified by the Hidden Markov Model (HMM) as bearing a posterior probability greater than 0.5 of being unaltered were subsequently reclassified as "normal" and were not included in the final aneuploidy delineation (Supplementary Figure 3I).

Cell subtype

We adopted the T cell annotations from the primary study but conducted additional sub-clustering for myeloid and cancer cell populations. Clustering resolution was established at 0.25 for cancer cells and 0.2 for myeloid cells, a strategy aimed at optimizing the delineation of cell groups. Guided by clustree, this approach carefully negotiates the balance between excessive fragmentation of cell clusters due to minor biological variations and the conflation of distinct cell types into overly broad groups. After re-clustering in subset cells, differential expression analysis were applied to identify differentially expressed genes in each cluster. The default Wilcoxon Rank Sum test was implemented due to computational limitation of our large sample number. The differentially expressed genes were then removed if false discovery rate (FDR) over 0.05 and ordered by their average log2 fold change. The difference in percent of cells expressing each gene were also calculated by Add_Pct_Diff from scCustomize.

Cell composition analysis

To evaluate the variance in cell-type composition between patient cohorts delineated by chromosome 2q status, we adopted a methodical approach. Initially, we visualized the distribution of cell clusters via stacked bar plots, categorizing them by their 2q status. Subsequently, we pinpointed specific cell subtypes of interest, guided by established background knowledge and by identifying clusters with visibly distinct proportions associated with 2q status. To assess the statistical significance of the differences in cell-type proportions, we conducted chi-

square tests. However, we recognized that the large sample sizes typical of single-cell data could undermine the chi-square test's efficacy, often yielding small p-values regardless of actual differences.

For more robust statistical validation, we utilized the scCODA (single-cell Compositional Data Analysis) framework, a Bayesian hierarchical model leveraging a Dirichlet-Multinomial distribution to mitigate uncertainties and inverse correlations within cell count data (27, 28). We implemented scCODA framework, specifically the package `pertpy` (v0.4.0), within Python (v3.11) to analyse T-cell populations, integrating data from the CELLxGENE database augmented with 2q status annotations (Supplementary Table 1A). Attempts to apply scCODA to myeloid cells were hindered as the conversion of re-clustered myeloid cells from a Seurat object to an AnnData object was impeded by persistent issues with the 'SaveH5Seurat' from SeuratDisk (v0.0.0.9021) (<https://github.com/mojaveazure/seurat-disk/issues/27>).

As an alternative, we applied the scDC (v0.1.0) available in R, a method utilizing bootstrap resampling to discern differential cell-type composition (29). We used 'scDC_noClustering' from scDC utilizing various confidence interval calculations—percentile, BCa, and multinomial—with a bootstrap value of 50. General linear models were then fitted to the data to visualize confidence intervals and calculate p-values, with the false discovery rate for scDC determined using the FDR method on the p-values obtained (Supplementary Table 1B).

IFN- γ and IFN score calculation

To quantify the interferon-gamma (IFN- γ) score, we employed a signature comprising *IFNG*, *CCR5*, *CXCL9*, *CXCL10*, *CXCL11*, *IDO1*, *PRF1*, *GZMA*, and *HLA-DRA* (30), utilizing 'AddModuleScore' from the Seurat. This approach calculates the mean expression level of each specified gene within individual cells of a cluster, which is then adjusted by subtracting the mean expression of a control gene set. The control genes are selected randomly from expression bins matched to the analysed genes, ensuring a balanced comparison (31). The resultant scores across clusters are aggregated to derive an overall IFN- γ score (Supplementary Figure 4 A).

Additionally, we derived an interferon score using a set of markers recognized for their roles in the interferon response pathway, namely *IFI27*, *IFIT3*, *OAS1*, *LY6E*, *IFI44L*, *MX1*, *SERPING1*, *IRF7*, and *C1QA*, following methodologies outlined in prior studies (32, 33). The process involves calculating the average expression of these IFN-related genes per Seurat cluster, alongside the standard deviation of expression for each gene within the clusters. The interferon score is computed using the formula: $(\sum((RE_subject - Mean_hc) / SD_hc))$, where *RE_subject* represents the expression level of IFN genes for each cluster, *Mean_hc* is the average expression of these genes across all clusters, and *SD_hc* is their respective standard deviations. This normalized IFN expression for each gene is then cumulated across clusters to establish a composite IFN score (Supplementary Figure 4 B-C).

Differentially Expressed Genes and Gene Set Enrichment Analysis

Differential gene expression analysis between chromosome amplification statuses was conducted utilizing the Seurat 'FindMarkers', paired with the MAST test—a method

tailored for single-cell transcriptomic data (34). In this analysis, the parameter `ident.1` was systematically assigned to represent the amplified chromosome group. Subsequent to identifying differentially expressed genes, we augmented the resulting dataset with a column quantifying the absolute disparity in the percentage of cells within each group that express a given gene.

For gene set enrichment analysis (GSEA) (v4.3.3), we employed the software available from the Broad Institute, focusing on pathway elucidation within Reactome and Gene Ontology's Biological Processes domains (35, 36). This was facilitated by employing the gene set matrices `c2.cp.reactome.v2023.2.Hs.symbols.gmt` and `c5.go.bp.v2023.2.Hs.symbols.gmt` (37). The analysis was conducted with the collapse method set to 'no_collapse', maintaining all other parameters at their default settings within the graphical user interface.

Intercellular interaction analysis

Cell-cell communication analysis were conducted by CellChat (v2.1.2), categorized according to chromosome arm status (38). Utilizing the human-specific CellChat database, we focused on ligand-receptor interaction pairs, excluding non-protein signalling database. This analysis identifies genes markedly overexpressed in each cell type, alongside the corresponding upregulated ligand-receptor interactions. Communication probabilities between cells were derived using a 'TriMean' method, enhancing the analysis by excluding interactions observed in fewer than 10 cells.

BISCUT identifies target genes

BISCUT, or Breakpoint Identification of Significant Cancer Undiscovered Targets, is an analytical method designed to pinpoint genomic regions under selective pressure through the examination of partial somatic copy number alterations (2). Following the protocol delineated in the authors' guide, employed BISCUT_preprocessing.py to create breakpoint files delineating telomeric and centromeric alterations for each chromosome arm from the TCGA purity-adjusted segmented copy number dataset. These files served as inputs for BISCUT_peak_finding.R, which isolates loci under selection, targeting a pre-defined set of interferon signalling genes comprising 637 unique identifiers. We extracted the genes located on 2q from all BISCUT_results output and correlated their gene expression with chr 2q copy number estimated by GISTIC2.0. False discovery rate was applied to adjust for multiple testing.

Results

Arm-level aneuploidy associates with Immune Phenotypes

To delineate the association between aneuploidy and immune response modulation, we firstly estimated arm-level aneuploidy from TCGA copy number segmentation data. We conducted linear regression analyses to assess the relationship between arm-level copy number alterations and immune features (i.e., *PRF1*, *MHCI*, *MHCII*, *GZMB*, *GNLY*, *CD274*, TMB, tertiary lymphoid structures (TLS), and immune cell infiltrations of T cells and Monocytes). Age, stage, sex, TMB, MSI score, and tumour subtype are incorporated as covariate to account for potential confounding effects (6, 39, 40). Chromosome arms that exhibited statistically significant association with the selected immune phenotypes were selected and plotted in Figure 1.

In ovarian cancer, tumours with chromosome arm 2p, 2q amplifications demonstrated a significant downregulation of all immune markers and lower expression of TLS, TMB and lower infiltrations of immune cells. It suggests chromosome 2 may have an important role in assisting immune evasion in ovarian cancer.

Lung cancer samples exhibited notable negative association on chromosome arms such as 22q. Where higher copy number corresponded to decreased expression of cytotoxic genes such as *PRF1*, *GZMB*, *GNLY*. Furthermore, MHC Type I and II genes along with tumour mutational burden showed a significant downregulation in the presence of gain on chromosome 15q, indicating a potential mechanism for immune evasion. Positive associations were observed on chromosome arms such as 9q and 9p, where higher copy number corresponded to increased expression of cytotoxic genes such as *GZMB*, *GNLY*, and *CD274*. Potentially associated with better response to immune therapy and prognosis. The infiltrations of T cells, B cells, and monocytes were also significantly lower in 15q and 22q amplification and higher in 9p and 9q amplification.

Chromosome 2q amplification in ovarian cancer associates with prognosis and therapy response

To further explore these preliminary findings, we utilized an independent high-grade serous ovarian cancer (HGSOC) cohort, which was compiled from patients treated at Hammersmith Hospital with a combination of surgery and chemotherapy. This dataset encompasses records from 119 patients. Among these, survival analysis were feasible for 115 patients and treatment response analysis were feasible for 41 patients. We also collected data on 52 lung cancer patients from the same hospital, all of which were treated with immunotherapy and had comprehensive survival information available. Given the relatively limited sample size for the lung cancer subgroup, we incorporated an extra set of lung cancer data with patients who had also been treated anti PD-1/PD-L1 blockers (41). This additional dataset features 240 patients with non-small cell lung cancer (NSCLC), all of whom underwent genomic profiling through IMPACT sequencing and whole exome sequencing (WES). This integration of datasets aims to solidify our understanding of the genomic factors influencing treatment outcomes in lung cancer, especially in the context of immunotherapy.

In our exploratory analysis, we delineated chromosomal aberrations with putative correlations to immune phenotypes. Therefore, we proceed only with chromosomes that have demonstrated significant ($FDR < 0.05$) association with immune features. We performed survival analysis using the Cox proportional hazards model with arm-level copy number alterations as variables of interest, while adjusting for age, sex, stage and TMB. Hazard ratios (HRs), along with 95% confidence intervals and p-values, were computed, and subsequent adjustment for multiple testing was applied using the false discovery rate method. The forest plot of hazard ratio demonstrates that 8p copy number increase associates with a significant decrease in hazard ratio of death, whereas 2q copy number increase significantly associates with an increase in hazard ratio. Suggesting that increase in 2q copy number corresponds to an increase in the risk of death. Interested in how these chromosome arms associates with key immune markers including *CD20* (marker of TLS), *CD4*,

Figure 1

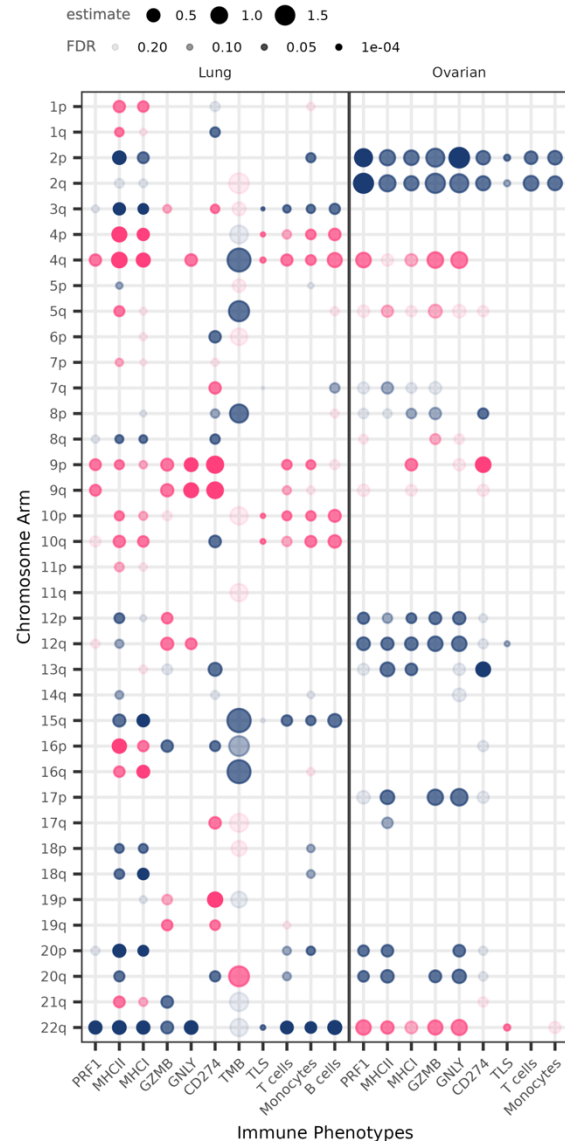
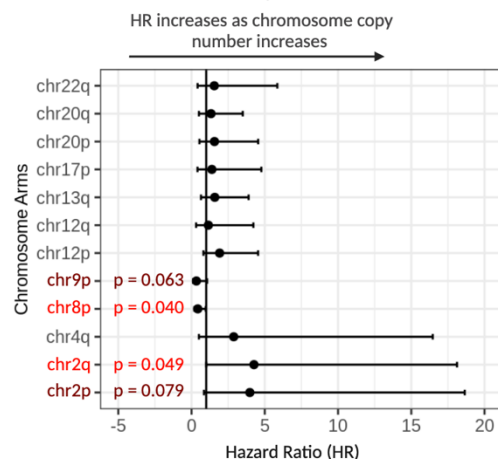


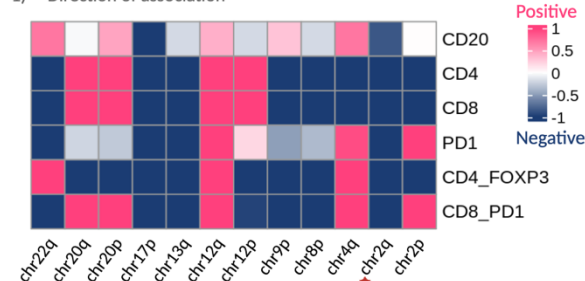
Figure 1 | Correlation of Chromosomal Arm-Level Copy Number Alterations with Immune Features in Lung and Ovarian Cancer. Linear regression on arm level CNA against selected immune features were performed. Age and stage are added as confounding variables in the formula. Dot size corresponding to the magnitude of the correlation while color gives the direction of the estimate, such that red is positive estimation and blue is negative estimation. Dot transparency represents levels of statistical significance calculated by false discovery rate. TMB = tumour mutational burden. TLS = tertiary lymphoid structure.

CD8, *PD1*, *CD4/FOXP3*, and *CD8/PD1*, we estimated the direction of association using linear regression and calculated the significance of the difference in FDR. The result was plotted using heatmap as in Figure 2B. Essentially speaking, 2q demonstrates the highest overall level of significance in association with immune markers (Figure 2B-2). With the direction of associations for all of the markers being negative, agreeing with our exploratory analysis in Figure 1.

Continuing on chromosome 2q, we plotted the Kaplan Meier survival diagram. Patients separated by their 2q status (amplification chr2q > 2 or wildtype chr2q ≤ 2). The 5-year overall survival rate was statistically different between patients with 2q amplification and those without ($p = 0.038$) (Figure 2C).

Figure 2**(A) Ovarian cancer survival forest plot****(B) Association between chromosome-arm amplification and immune gene expression**

1) Direction of association



2) Strength of association

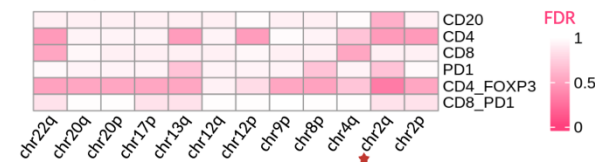
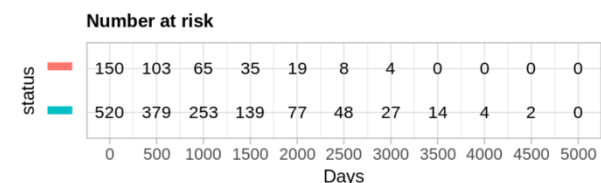
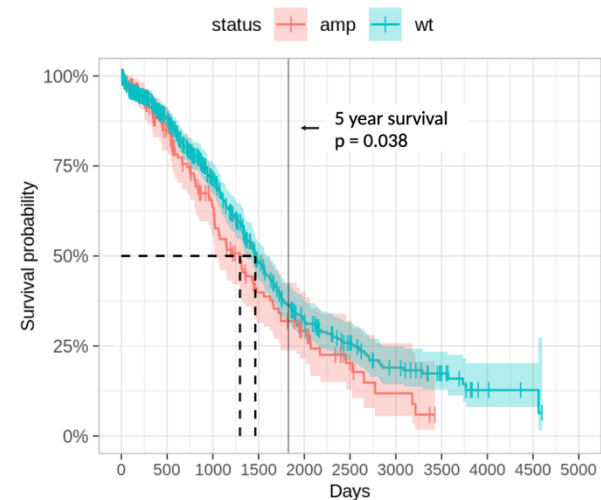
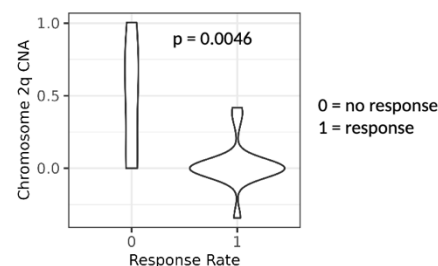
**(C) Kaplan-Meier Curve for Ovarian Cancer**OS, *hh_ova* + *tcga_ova* (670), *surv days* < 5000**(D) chr2q copy number by therapy response**

Figure 2 | Immuno- and Clinical- Association with Chromosome Arm Copy Number in Ovarian Cancer. (A) Forest plot illustrating the impact of arm-level copy number alterations on patient survival, analysed using a proportional hazards regression model (coxph). Hazard ratios are presented with their 95% confidence intervals. Chromosomal arms with a false discovery rate (FDR) less than 0.25 are highlighted in red, with a lighter shade for $p < 0.05$ and darker shade for $p > 0.05$. (B) Heatmap showcasing the relationship between chromosome arm amplification and immune gene expression, with association direction inferred through linear regression and association strength through Wilcoxon test, with multiple testing correction by false discovery rate. (C) Kaplan-Meier survival curve segregating ovarian cancer patients based on chromosome 2q status (amplified or wildtype), a vertical line at 1826 days marks the study's 5-year overall survival rate ($p = 0.035$), indicating differential survival rates. Survival differences were evaluated using the log-rank test. (D) Violin plot depicting the difference in distribution of chromosome 2q copy number in patients with treatment response (1) vs without treatment response (0), significance of the difference is estimated by Wilcoxon test.

Lastly, we calculated the distribution of 2q copy number for patients with or without chemotherapy treatment response (Figure 2D). The violin plot demonstrates that for patients without treatment response their 2q copy number distribution is mostly amplified, without any deletion events. Whereas for patients with treatment response the majority of their 2q copy number change centred around 0 (diploid state) with a few amplifications and deletions (p -value = 0.0046).

For lung cancer, no chromosome arms attained statistical significance, with chromosome 15q achieving a p -value of 0.072 for Hazard Ratio and 0.0644 for comparison of chromosome 15q copy number distribution between treatment response and no-response group. The Kaplan Meier plot differentiating chromosome 15q deletion ($cn < 2$), wildtype (diploid, $cn = 2$), and amplification ($cn > 2$) demonstrates a p -value of 0.03 for progression free survival. Overall survival data was not available for majority

of samples therefore PFS was used as surrogate. (Supplementary Figure 5 A-D).

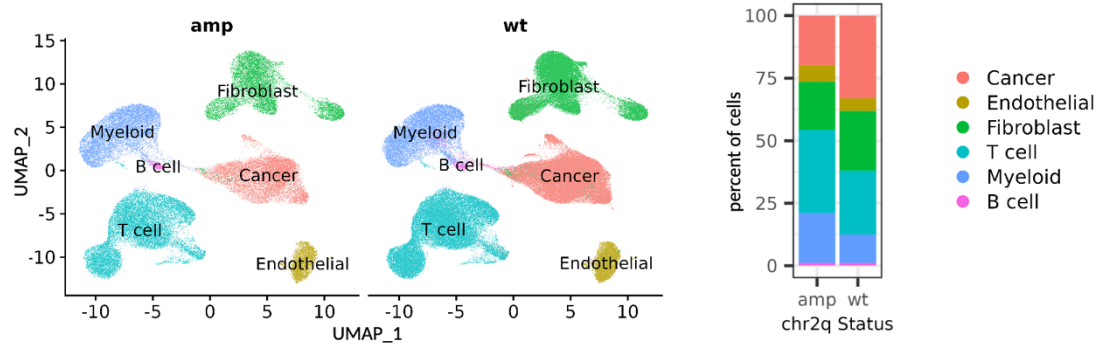
Immunosuppressive phenotype observed in cancer cells with amplified 2q

Following the establishment of a correlation between 2q amplification in ovarian cancer and diminished immune signatures, lower immune cell infiltrations, suboptimal treatment response, and poorer survival outcomes, we sought to elucidate the molecular underpinnings of these clinical manifestations. Subsequently, we categorized patients based on their 2q status.

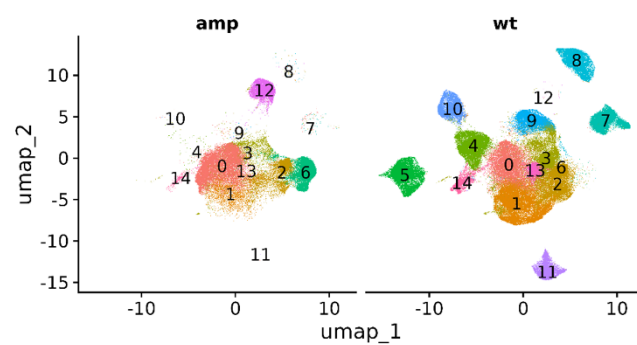
In delineating cell type discrepancies between the amplified and wild-type cohorts, we utilized Uniform Manifold Approximation and Projection (UMAP) to visualize the major cell types including cancer, endothelial, fibroblast, T cell, B cell and Myeloid cells and constructed a stacked bar plot to quantify cell type proportions (Figure 3A). This preliminary analysis showed that there is a

Figure 3

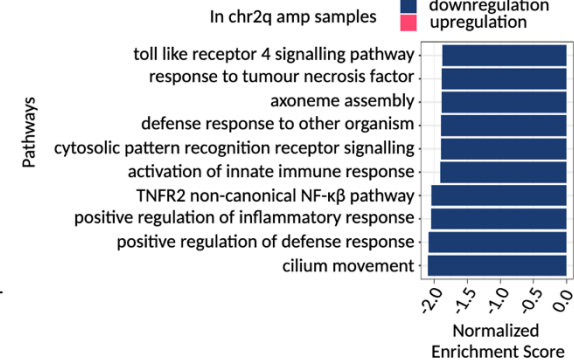
(A) Panel plot ovarian cancer single cell sequencing data UMAP diagram and stacked bar plot by major cell type



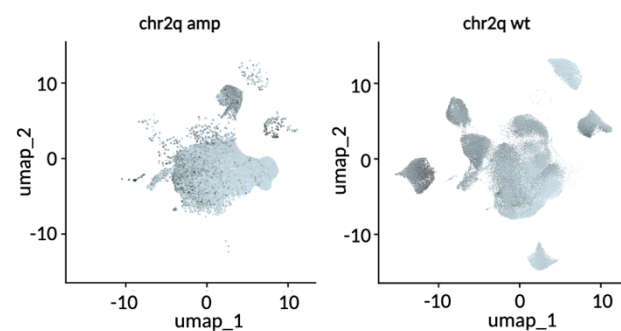
(B) UMAP projection of cancer cell clusters



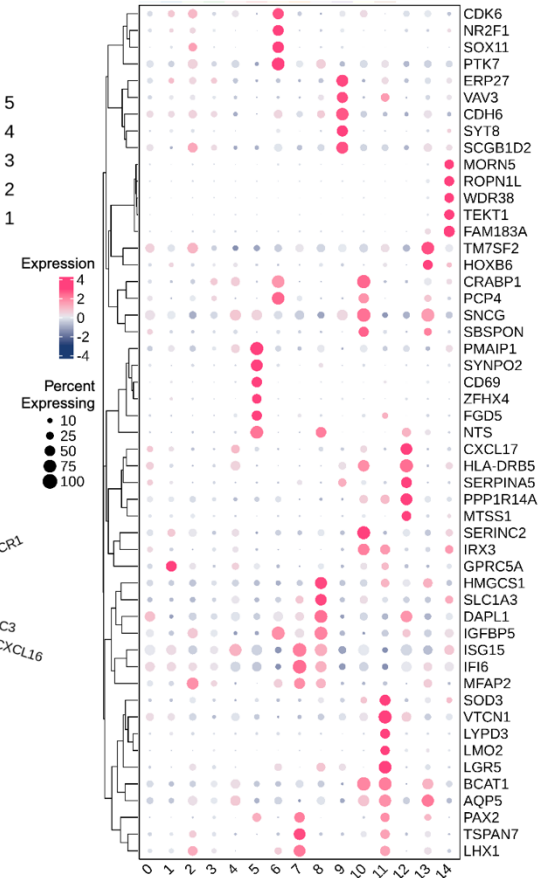
(C) two-side bar plot of GSEA



(D) Feature plot of IFN-γ score



(E) Clustered Dot Plot of top differentially expressed genes



(F) Cell Chat Chord Plot for CCL, CXCL, SPP1, and COMPLEMENT pathways

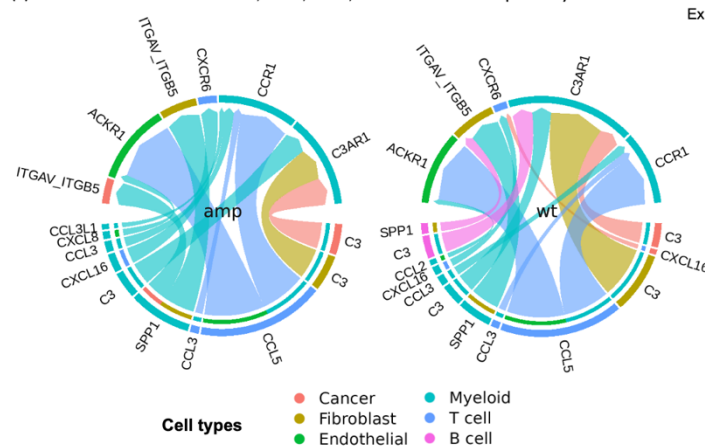


Figure 3 | Single cell RNA sequencing analysis (A) UMAP projection of single-cell sequencing data delineating individual cells and colors indicating cell type; accompanied by a stacked bar plot showing the proportion of cell types in chr2q amplified and wildtype groups. (B) UMAP projection of cancer cell clusters, clusters calculated with resolution of 0.25. (C) Gene expression differences between chr2q subsets identified via MAST algorithm, suitable for scRNA-seq data. (D) Feature plot displaying IFN- γ scores in cancer cells, computed using Seurat's AddModuleScore function across IFN- γ gene sets. (E) Clustered Dot Plot of top differentially expressed genes in each cancer cluster calculated by wilcox test, top 5 genes from each cluster taken (F) Chord diagrams illustrating intercellular communication for selected pathways between chr2q amplified and wildtype cells.

pronounced reduction of myeloid lineage cell counts in amplified chr2q cohort compared to wild type. Whereas the T cell subtype was rather counter-intuitively equal. We first conducted a differential gene expression analysis of cancer cells with and without chromosome 2q amplification, utilizing the MAST algorithm. Subsequent GSEA, leveraging reactome and the gene ontology biological process database, failed to identify any statistically significant upregulated pathways in cells with 2q amplification under the stringent threshold of $FDR < 0.05$. Contrarily, we observed a marked downregulation of numerous immune pathways in 2q-amplified cells, including the toll-like receptor 4 signalling pathway, TNFR2 non-canonical NF- κ B pathway, and processes integral to the tumor necrosis factor response, innate immune activation, and the promotion of inflammatory responses. These findings collectively suggest an immune tolerance phenotype, indicative of immune evasion mechanisms in cancer cells harbouring 2q amplification. Given the innate immunity focus of these downregulated pathways, we were prompted to examine the interferon signature within these cancer cells more closely. Our analysis, illustrated in Figure 3C and Supplementary Figure 4A-C, encompassed both general interferon and interferon- γ scores. Linear regression analysis underscored a negative correlation between the interferon- γ score and 2q copy number, indicating that an increased copy number correlates with reduced IFN- γ score. This pattern, consistent across general interferon scores, intimates that genes within 2q, when amplified, may suppress interferon signalling pathways, thereby dampening the immune response to tumor cells and facilitating their evasion from immune detection and destruction.

To further elucidate the role of 2q amplification in altering cellular communication, we employed CellChat analysis, identifying a diverse array of 50 pathways—both shared and exclusive (SELL and BAFF in the amplified subset; PLAUR and CADM in the wild-type)—implicated in oncogenesis. This analysis revealed significant differences in 10 shared pathways between the cohorts, notably MK, FN1, PERIOSTIN, ANNEXIN, PTN, APoE, COMPLEMENT, SPP1, CXCL, and CCL, the last four of which are closely linked to mechanisms of immune evasion (Supplementary Table 2). Notably, in the 2q amplified cohort, there was a distinct absence of CXCL-mediated interactions with cancer cells, a pathway crucial for immune communication. In contrast, this group uniquely showed SPP1 signalling directly between cancer cells and myeloid cells, diverging from the wild-type's SPP1 communication pattern involving B cells, fibroblasts, and myeloid cells. A deeper investigation into specific ligand-receptor interactions revealed a significant absence of CXCL16-CXCR6 communication in the 2q amplified group. This axis has been previously recognized for its role in recruiting activated T cells to tumor sites, thereby enhancing the immune response to therapies such as combined radio-immunotherapy (42). The lack of CXCL16-CXCR6 interaction in the 2q amplified samples imply a compromised ability to recruit T lymphocytes, facilitating a pathway for cancer cells to evade immune surveillance. Consequently, our focus shifts towards examining the T cell subset more closely, aiming to uncover disparities in composition and communication between the amplified and wild-type cohorts, which could further

illuminate the mechanisms of immune evasion driven by 2q amplification.

Downregulation of Interferon pathways in T cells

Figure 4

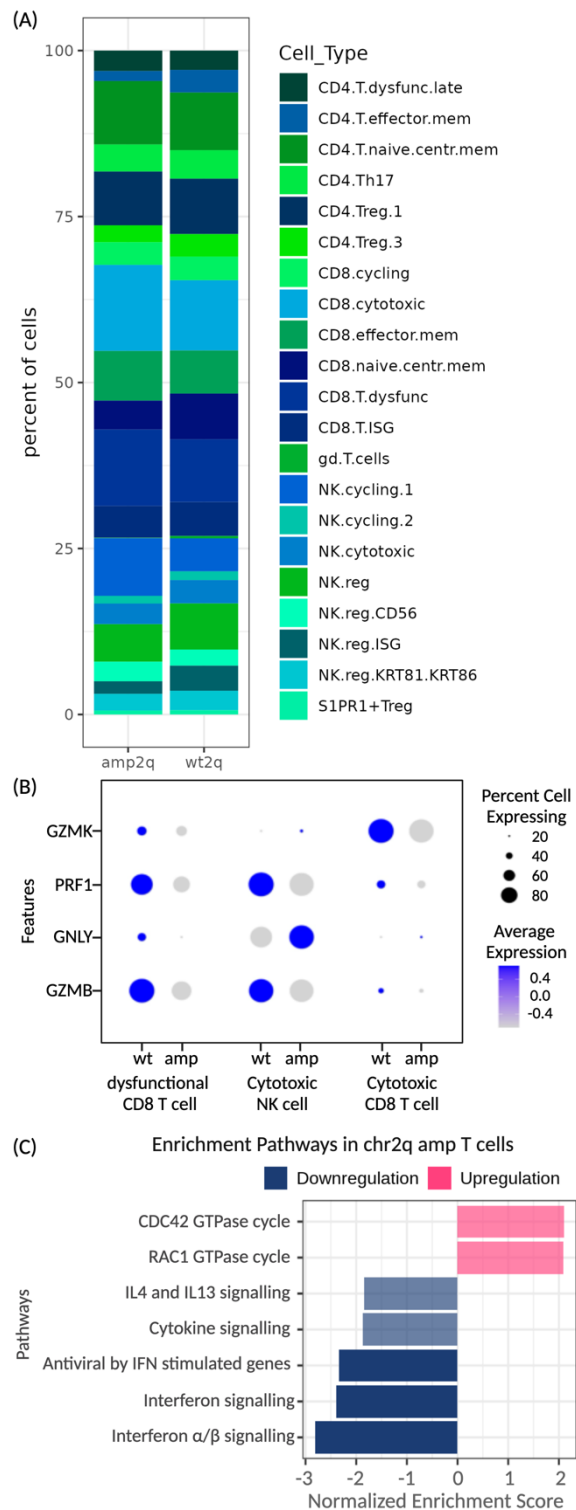


Figure 4 | Analysis of T cells. (A) Stacked bar plot showing the proportional representation of T cell subsets within chr2q amp and wt conditions. (B) Expression of selected immune markers in dysfunctional CD8, cytotoxic NK, and cytotoxic CD8 cells split by chromosome 2q status (wild type vs amplified). (C) Two way bar chart highlighting the top enriched Reactome pathways in chr2q amp T cells, with directional bars indicating upregulation (red) and downregulation (blue) of pathways. Pathways with $FDR > 0.05$ but < 0.20 has lighter color.

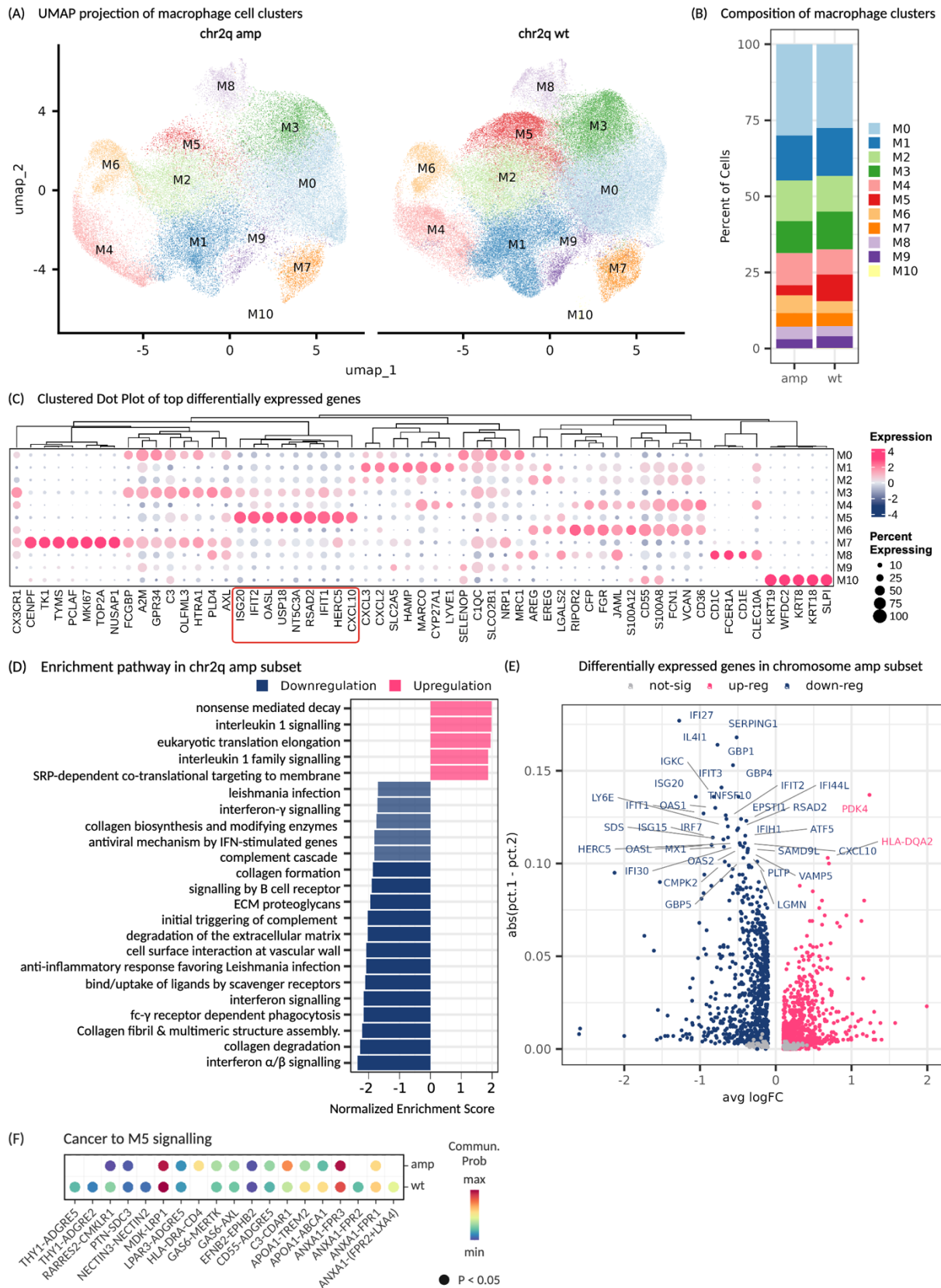
Figure 5

Figure 5 | Multifaceted Analysis of Macrophage Cells and Pathways in chr2q Amplified and Wildtype Cells (A) UMAP plots representing single-cell RNA sequencing (scRNAseq) data to show the clustering of macrophage cells in samples with chromosome 2q amplification (chr2q amp) and wildtype (chr2q wt). Each dot represents an individual cell and colored by cluster. (B) Stacked bar plot of the distribution of cells (%) in different macrophage cell clusters of amplified and wild type cohort. (C) Dot plot of top differentially expressed genes in each cluster, genes pre-filtered by adjusted p-value < 0.05 , percent of cell expressing > 0.30 , average log2 fold change > 1 , then top 5 genes from each cluster is selected as markers for each cluster. (D) Enrichment pathway in chromosome 2q amplification subset by gene set enrichment analysis using Reactome database. (E) Differentially expressed genes in chromosome 2q amplification subset. (F) Bubble Plot of CellChat identified dysfunctional signaling ligand-receptor pairs between cancer cells to macrophage M5 cells.

In our investigation, we concentrated on a comprehensive analysis of T cells. We began by determining the percentages of various T cell subtypes across two groups (Figure 4A), revealing a noteworthy increase in cytotoxic CD8 T cells, dysfunctional CD8 T cells, and cytotoxic NK cells among the cohort with 2q amplification. This observation was surprising given the established link between 2q amplification and an immunosuppressive phenotype in previous studies.

After integrating T cell data with 2q copy number data and predicated on 2q amplification status, scCODA analysis revealed no significant variance in cell subtype presence at an FDR threshold of 0.20.

Subsequent to identifying T cell subtype enrichments, we mapped the expression of T-cell effector molecules (*GZMK*, *PRF1*, *GNLY*, *GZMB*) in these subtypes, delineating them by 2q status (Figure 4B). We found that dysfunctional CD8 cells from patients with 2q amplification showed reduced expression of these markers, whereas cytotoxic NK and CD8 T cells experienced a decrease in *GZMK*, *PRF1*, and *GZMB* but an uptick in *GNLY* expression in 2q amplified tumours.

Further, utilizing the MAST algorithm for differential gene expression in single-cell data and conducting Reactome enrichment analysis, we noted significant downregulation in interferon signalling, especially IFN- α/β pathways, and decreases in cytokine and IL4/IL13 signalling pathways (Figure 4C). Considering myeloid cells are primary producers of Type I interferons (IFN-alpha and IFN-beta) in response to infections, these findings indicate a possible alteration in myeloid cell function, potentially leading to the emergence of an immunosuppressive phenotype.

Reduction of M5 macrophage seen together with suppression of IFN signaling in patients with chromosome 2q amplification

For an in-depth exploration, we isolated myeloid-lineage cells, opting for a distinct approach from the one used for T cells by not relying on inherent clustering. Instead, we applied 'FindClusters' from Seurat, allowing us to derive our own clustering schema. The results of this process were visualized in a UMAP projection, showcasing the myeloid cell clusters (Figure 5A), alongside the composition of these clusters expressed as percentages (Figure 5B). Differential cell composition analysis demonstrated a particularly noteworthy finding was the reduction of the M5 cluster in patients exhibiting 2q amplification ($p = 2.07e-09$) (Supplementary Table 1B). Further investigation through differential gene expression analysis between the clusters (Figure 5C) revealed that the most significantly altered genes in cluster M5 included *ISG20*, *IFIT2*, *OASL*, *USP18*, *NT5C3A*, *RSAD2*, *HERC5*, and *CXCL10*. The majority of these genes are known for their roles in interferon-stimulated gene expression and the broader sphere of immune regulation. It was observed that a significant portion of these genes, specifically *ISG20*, *IFIT2*, *RSAD2*, *HERC5*, and *CXCL10*, were considerably downregulated in patients with 2q amplification (Figure 5E). Delving deeper into the enrichment pathways within myeloid cells from patients with 2q amplification, we discovered a significant suppression of interferon signalling pathways, particularly those related to IFN- α/β and IFN- γ , as well as the Fc gamma receptors (Fc γ Rs) dependent phagocytosis activity within the myeloid cell population of

these patients (Figure 5D). This decrease in Type I IFN signalling corroborates with earlier observations concerning the pattern of IFN- γ score in cancer cells. Additionally, we identified downregulation in other vital pathways, such as complement signalling—from its initiation to the subsequent cascade—and collagen degradation activities. The suppression of complement signalling is consistent with the immunosuppressive phenotype previously associated with 2q amplification. The observed decrease in collagen degradation activity hints at an increased collagen density, leading to higher tissue stiffness. This phenomenon likely contributes to the impediment of T cell migration speed and fosters CD8+ T cell exhaustion, thus underlining a complex network of biological processes influenced by 2q amplification that fosters an environment conducive to immune evasion (43).

BISCUT complements known targets on 2q

We utilised the BISCUT algorithm to uncover potential immune regulator and therapeutic targets on chromosome 2q amplification. We downloaded TCGA pan-cancer dataset and applied the BISCUT pipeline to it as suggested by original paper (2). From the result we isolated genes appearing on 2q and correlates these gene's expression with 2q copy number. From those, we identified CFLAR (estimate = 0.52, FDR = $2.49e-14$) and ATG9A (estimate = 0.66, FDR = $2.43e-31$) as having statistically significant correlation with 2q copy number.

Discussion

Key findings

In this study, we uncovered a novel link between 2q amplification and immune suppressive phenotype in ovarian cancer, contributing a new angle of understanding how genomics underpins immune evasion. This amplification was associated with downregulation of key immune markers including *PRF1*, *MHCI*, *MHCII*, *GZMB*, *GNLY*, *CD274* and a decrease in T cell and monocyte density alongside TLS score in patients from TCGA ovarian cancer dataset. We validated our discovery by demonstrating that 2q amplification correlates with downregulation of *CD20*, *CD4*, *CD8*, *PD1* in HGSOC data derived from Hammersmith hospital. Moreover, we proved that 2q amplification is associated with worse patient prognosis ($p = 0.038$) by survival analysis comparing patients with 2q amplification and those without using survival data from both TCGA and Hammersmith Hospital.

Utilising single cell sequencing data from 42 HGSOC patients, we pinpoints the molecular mechanism of 2q amplification in tumour microenvironment and tumorigenesis. Specifically, we identified a novel relationship between reduction of interferon pathways in patients with 2q amplification. As demonstrated by the statistically significant decrement in the IFN-gamma score within 2q amplified cancer cells ($p < 2.2e-16$), a downregulation of type I IFN pathway in T cell, and downtrend of both type I and II IFN pathway in myeloid cells. This cross-cellular IFN pathway suppression hints at a widespread genomic influence exerted by 2q amplification on immune landscape. Type I interferons can activate myeloid cells (like dendritic cells) to produce antiviral responses and present antigens to T cells more effectively. Whereas IFN-gamma, is crucial for "licensing" dendritic

cells to prime T cells, playing a vital role in T cell activation, differentiation, and function (44).

Wishing to pinpoint the exact cell type(s) responsible for the dysregulation observed, we delve deeper into macrophage and T cells in single-cell level, comparing the proportion of cell subtypes in patients with amplified or wildtype 2q. We identified a macrophage cell cluster, denoted as M5, expressing a suite of interferon-stimulated genes (*ISG20*, *IFIT2*, *OASL*, *USP18*) as their top differentially expressed entities. Moreover, M5 also highly expresses *CXCL10* as top differentially expressed gene markers, known to regulate tumour immune microenvironment in ovarian cancer, thus exerting anti-tumour function. Specifically *CXCL10* suppresses angiogenesis and activates immune killing response and therefore has been proposed as a potential direction to improve immunotherapy effectiveness in ovarian cancer (45). Furthermore, utilising BISCUT we identified *ATG9A* as a novel marker associating with 2q amplification.

ATG9A, autophagy, and IFN-signaling

Autophagy is the cellular process for degrading and recycling components to maintain cell stability and survival. *ATG9A*, situated on chromosome 2q35, encodes a protein crucial to the autophagy pathway, essential for cellular survival and balance. This gene orchestrates lipid transport to autophagosomes, initiating autophagy. It dynamically shuttles among various cell compartments, such as the trans-Golgi network, endosomes, and plasma membrane and is essential for autophagy initiation (46, 47). *ATG9A* exhibits varied roles in different cancers; it's overexpression is linked to worse outcomes in oral squamous cell carcinoma and progression of benign prostatic hyperplasia via elevated autophagy level (47). Whereas its inhibition relates to therapy resistance against Trastuzumab in HER2+ breast cancer and reduced cancerous traits in triple-negative breast cancer. Interestingly, both HER2+ and invasive ductal carcinoma tissues show lower *ATG9A* expression compared to normal tissue, underscoring its complex function in cancer progression and patient prognosis (48). Biologically, the *ATG9A* participates in the cGAS-cGAMP-STING pathway by regulating autophagy (Figure 6), and monocyte is the primary source at which STING pathway is stimulated to produce type 1 IFN. This pathway is activated upon the detection of DNA fragments in the cytosol by the cGAS protein, which then initiates the synthesis of cGAMP. SLC19A1 facilitates the binding of cGAMP to the STING dimer, thereby transporting cGAMP from the extracellular space into the cytosol and converting STING to its active conformation. The TRAP β protein assists in transporting the activated STING from the endoplasmic reticulum to the Golgi apparatus. There, STING binds to TBK1 through phosphorylation. The STING-TBK1 complex can then phosphorylate the IRF3 transcription factor or NF- κ B, which are subsequently able to translocate to the cell nucleus, bind to the IFN-1 promoter, and induce the expression of IFN-1 associated genes (49). During this process, *ATG9A* plays a crucial role in regulating the localization of the STING protein from the Golgi apparatus to the punctate structure during the assembly of the STING-TBK1 complex, and the loss of *ATG9A* greatly enhances this assembly. Similarly, *ATG9A* loss also enhances dsDNA-induced IRF3

phosphorylation, thus enhancing the downstream transcription of IFN- β , IL-6 and *CXCL10* proteins. Given the reciprocal relationship between *ATG9A* expression and STING-TBK1 complex assembly. It is possible that the increased expression of *ATG9A* might negatively influenced the IRF3 phosphorylation and IFN-1 production. In our study we observed an upregulation of *ATG9A* expression due to chromosome 2q amplification (as *ATG9A* is located on 2q35) along with a downregulation of IFN-signalling pathway and general immune evasion/suppression phenotype. Type I IFN exerts its anti-tumor effects through multiple mechanisms. Includes promoting dendritic cell (DC) maturation, promoting surface expression of MHC class I molecules, preventing degradation of phagocytized tumor antigens, and improve lymph node homing and migration of DCs via upregulation of chemokines such as CCR7, Th-1, and MIP-3 β . All of which potentiates the activation of CD8+ T cells. In addition *CXCL10* and *CXCL9* secretion were also upregulated by type I IFN, and they are crucial for the recruitment and infiltration of cytotoxic T lymphocytes into tumor sites. *CXCL10* was observed to be downregulated in macrophages from patients with chromosome 2q amplification in our study (49).

Moreover, studies have also found that STING-deficiency is associated with weakened T cell antitumour immune function and a reduction of immune checkpoint inhibitor efficacy (49, 50). This is similarly observed in NK cells, such that STING-deficiency decreases immature NK cells with self-renewal capacity since STING pathway positively regulates NK cell antitumour immunity and plays a role in maintaining TCF-1+ NK cell reservoir (51). We therefore speculates that the immune evasion mechanism and immunosuppressive phenotype observed in patients with chromosome 2q amplification is at least partly a result of reduction in cGAS-cGAMP-STING pathway and specifically the assembly of STING and TBK1 complex and IFN production (52, 53).

Limitations, challenges, and future directions

Currently there are many technological and computational limitations in studying the role of aneuploidy in cancer. In our analysis, we opted for the FACETS package over ABSOLUTE for chromosome segmentation, as ABSOLUTE is no longer supported by the Broad Institute. FACETS, requiring paired tumor and normal BAM files, presented difficulties since normal samples were unavailable for majority of our lung samples, leading to potential biases, especially given the gender-specific normal samples obtained from ovarian cancer compared to our mixed-gender lung cancer cohort. Additionally, the reliance on certain computational tools, such as CellChat, GISTIC2, and Seurat, which have their limitations and dependencies on parameter settings, necessitates cautious interpretation of results.

Our cell type annotation for ovarian cancer single-cell sequencing, particularly for T cells, may lack precision due to incomplete manual labelling inherited from original study, prompting us to develop our clustering for myeloid cell analysis. Furthermore, prevailing analytical tools for differential cell type composition, such as scCODA and scDC, may yield incongruent outcomes, casting doubt on the existence of genuine statistical disparities between two cell populations (Supplementary Table) (28).

While this study offers insights into arm-level copy number alterations (CNAs), particularly on 2q, it might not capture the full spectrum of genomic influences on tumor immunology and patient outcomes. Due to time limitation, the focused analysis on specific cell types like T cells and macrophages sheds light on some aspects but may not encompass the tumor microenvironment's complexity, including stromal and B cells, non-cellular elements, and certain intercellular communications.

Looking ahead, our research agenda is set to further dissect the impact of aneuploidy on the tumor microenvironment. In lung cancer, we wish to proceed our analysis of chromosome 15q in single cell sequencing data. In ovarian cancer, we plan to initiate our future studies by exploring the influence of aneuploidy on stromal and B cells, thus aiming to provide a more nuanced understanding of the tumoral landscape in ovarian cancer. This stems from emerging evidence suggesting a pivotal interaction between B cells and T cells mediated by the STING pathway, which warrants a deeper investigation (54, 55). Building upon the hypothesis that chromosome 2q amplification may disrupt the autophagy pathway—potentially through the upregulation of ATG9A—and subsequently affect IFN signalling and immune cell infiltration, we intend to leverage CRISPR-Cas9 screening techniques to precisely identify the culpable genes on chromosome arm 2q (53). This approach will not only clarify the mechanistic underpinnings, but also set the stage for developing therapeutic strategies that target these genomic aberrations. Understanding the role of chromosome 2q in immune evasion in ovarian cancer could revolutionize the therapeutic landscape, potentially transforming aneuploidy into a targetable feature that can be exploited to enhance the effectiveness of immunotherapies.

Conclusions

Chromosome aneuploidy, a ubiquitous feature across cancers, has historically been considered a secondary, non-driving event in tumorigenesis due to the vast number of genes affected by the amplification or deletion of entire chromosomes or chromosome arms, complicating the identification of specific genes that drive cancer progression. Prior research has primarily concentrated on delineating the relationship between aneuploidy and patient prognosis. Our investigation advances this understanding by examining the interplay between cancer aneuploidy and immune function. We have unveiled the previously unrecognized role of chromosome 2q amplification in promoting immune evasion within the context of ovarian cancer. Furthermore, our methodological approach substantiates the feasibility of investigating aneuploidy as a factor in immune evasion and, by extension, cancer immunology. Our exploratory analysis in lung cancer supports this perspective, suggesting that amplification of the 15q chromosomal arm may be linked to both immune response and patient survival. Leveraging the BISCUT analytical tool has allowed us to identify promising targets and pathways for further study, which may lead to the development of new therapeutic interventions that disrupt the pathways influenced by aneuploidy.

Codes and Data Availability

All the codes used in this study is available on GitHub page (<https://github.com/PollyHung/MRes.Project.1.git>). TCGA data were available on (<https://xenabrowser.net/>) and processed Hammersmith Hospital dataset were available along with the codes on the same GitHub page. Single Cell RNA sequencing data were available from GEO database under the succession number GSE180661 or from CELLxGENE database with this link (<https://cellxgene.cziscience.com/collections/4796c91c-9d8f-4692-be43-347b1727f9d8>). And corresponding patient information were available from cBioPortal (https://www.cbioportal.org/study/summary?id=msk_spec_trum_tme_2022).

Abbreviations

CNA, Copy Number Alteration; CNV, Copy Number Variation; ACNA, Aneuploidy Copy Number Alteration; SCNA, Somatic Copy Number Alteration; TCGA, The Cancer Genome Atlas; GSEA, Gene Set Enrichment Analysis; BISCUT, Breakpoint Identification of Significant Cancer Undiscovered Targets; HGSOC, High-Grade Serous Ovarian Cancer; LUAD, Lung Adenocarcinoma; LUSC, Lung Squamous Cell Carcinoma; IMPACT, Integrated Mutation Profiling of Actionable Cancer Targets; WES, Whole-Exome Sequencing; BAM, Binary Alignment Map; SNP, Single Nucleotide Polymorphism; UMI, Unique Molecular Identifier; PCA, Principal Component Analysis; STING, Stimulator of Interferon Genes; TBK1, TANK-Binding Kinase 1; IRF3, Interferon Regulatory Factor 3; IFN, Interferon; IFNG, Interferon Gamma; CCR5, C-C Chemokine Receptor Type 5; CXCL9, C-X-C Motif Chemokine Ligand 9; IDO1, Indoleamine 2,3-Dioxygenase 1; GZMA, Granzyme A; HLA, Human Leukocyte Antigen; TLS, Tertiary Lymphoid Structures; PD-L1, Programmed Death-Ligand 1; GNLY, Granulysin; GZMB, Granzyme B; PRF1, Perforin 1; FDR, False Discovery Rate; DC, Dendritic Cell; MIP-3 β , Macrophage Inflammatory Protein 3 Beta; CXCL10, C-X-C Motif Chemokine Ligand 10; TCF-1, T cell factor 1; NK, Natural Killer; cGAS, Cyclic GMP-AMP Synthase; cGAMP, Cyclic GMP-AMP; LAIR1, Leukocyte-Associated Immunoglobulin-Like Receptor 1; IFN- β , Interferon Beta; IL-6, Interleukin 6; CCR7, C-C Chemokine Receptor Type 7; Th-1, T Helper Cell Type 1; CXCL16, C-X-C Motif Chemokine Ligand 16; CXCR6, C-X-C Motif Chemokine Receptor 6; scCODA, single-cell Compositional Data Analysis; scDC, single cell differential composition; ATG9A, Autophagy-related protein 9A .

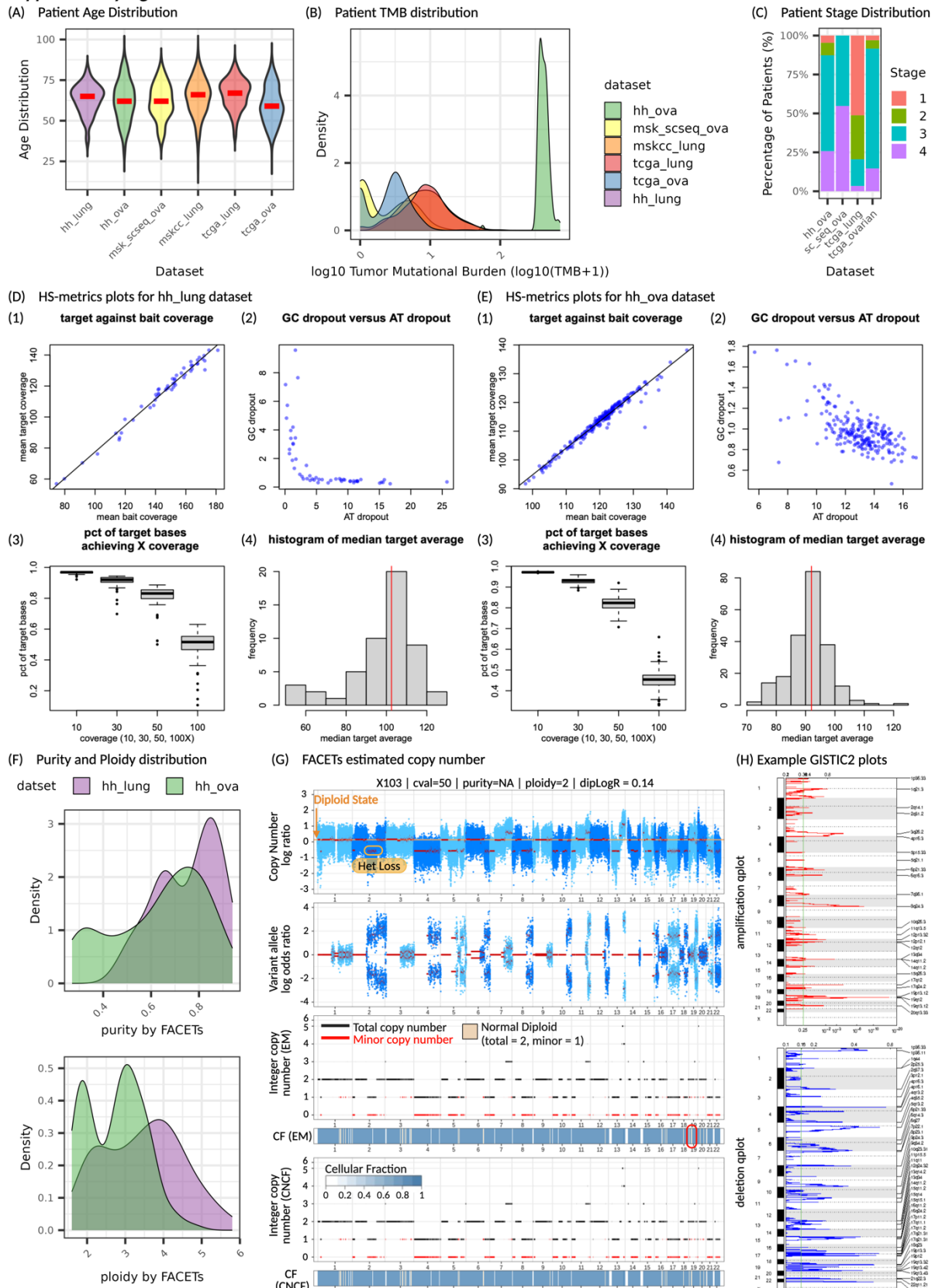
References

1. U. Ben-David, A. Amon, Context is everything: aneuploidy in cancer. *Nature Reviews Genetics* **21**, 44-62 (2020).
2. J. Shih *et al.*, Cancer aneuploidies are shaped primarily by effects on tumour fitness. *Nature* **619**, 793-800 (2023).
3. A. M. Taylor *et al.*, Genomic and Functional Approaches to Understanding Cancer Aneuploidy. *Cancer Cell* **33**, 676-689.e673 (2018).
4. M. Du, S. Zhang, X. Liu, C. Xu, X. Zhang, Ploidy Status of Ovarian Cancer Cell Lines and Their Association with Gene Expression Profiles. *Biomolecules* **13**, (2023).
5. M. Du *et al.*, Aneuploid serves as a prognostic marker and favors immunosuppressive microenvironment in ovarian cancer. *Journal of Ovarian Research* **17**, 30 (2024).
6. O. Molina, M. A. Abad, F. Solé, P. Menéndez, Aneuploidy in Cancer: Lessons from Acute Lymphoblastic Leukemia. *Trends Cancer* **7**, 37-47 (2021).
7. K. Salmina *et al.*, The Cancer Aneuploidy Paradox: In the Light of Evolution. *Genes* **10**, 83 (2019).
8. E. V. Watson *et al.*, Chromosome evolution screens recapitulate tissue-specific tumor aneuploidy patterns. *Nature Genetics*, (2024).
9. C. Brancolini, L. Iuliano, Proteotoxic Stress and Cell Death in Cancer Cells. *Cancers (Basel)* **12**, (2020).
10. J. Huang *et al.*, Worldwide Burden, Risk Factors, and Temporal Trends of Ovarian Cancer: A Global Study. *Cancers (Basel)* **14**, (2022).
11. A. Leiter, R. R. Veluswamy, J. P. Wisnivesky, The global burden of lung cancer: current status and future trends. *Nature Reviews Clinical Oncology* **20**, 624-639 (2023).
12. C. M. Rudin, E. Brambilla, C. Faivre-Finn, J. Sage, Small-cell lung cancer. *Nature Reviews Disease Primers* **7**, 3 (2021).
13. D. Bell *et al.*, Integrated genomic analyses of ovarian carcinoma. *Nature* **474**, 609-615 (2011).
14. U. A. Matulonis *et al.*, Ovarian cancer. *Nature Reviews Disease Primers* **2**, 16061 (2016).
15. U. Testa, E. Pelosi, G. Castelli, Molecular characterization of lung adenocarcinoma combining whole exome sequencing, copy number analysis and gene expression profiling. *Expert Review of Molecular Diagnostics* **22**, 77-100 (2022).
16. M. M. Bjaanaes *et al.*, Whole genome copy number analyses reveal a highly aberrant genome in TP53 mutant lung adenocarcinoma tumors. *BMC Cancer* **21**, 1089 (2021).
17. H. Rizvi *et al.*, Molecular Determinants of Response to Anti-Programmed Cell Death (PD)-1 and Anti-Programmed Death-Ligand 1 (PD-L1) Blockade in Patients With Non-Small-Cell Lung Cancer Profiled With Targeted Next-Generation Sequencing. *Journal of Clinical Oncology* **36**, 633-641 (2018).
18. R. Shen, V. E. Seshan, FACETS: allele-specific copy number and clonal heterogeneity analysis tool for high-throughput DNA sequencing. *Nucleic Acids Res* **44**, e131 (2016).
19. A. Arora, R. Shen, V. E. Seshan, in *Variant Calling: Methods and Protocols*, C. Ng, S. Pisuoglio, Eds. (Springer US, New York, NY, 2022), pp. 89-105.
20. C. H. Mermel *et al.*, GISTIC2.0 facilitates sensitive and confident localization of the targets of focal somatic copy-number alteration in human cancers. *Genome Biology* **12**, R41 (2011).
21. K. Chatsirisupachai, T. Lesluyes, L. Paraoan, P. Van Loo, J. P. de Magalhães, An integrative analysis of the age-associated multi-omic landscape across cancers. *Nature Communications* **12**, 2345 (2021).
22. R. Banichereau *et al.*, Molecular determinants of response to PD-L1 blockade across tumor types. *Nature Communications* **12**, 3969 (2021).
23. R. Cabrita *et al.*, Tertiary lymphoid structures improve immunotherapy and survival in melanoma. *Nature* **577**, 561-565 (2020).
24. E. Becht *et al.*, Estimating the population abundance of tissue-infiltrating immune and stromal cell populations using gene expression. *Genome Biology* **17**, 218 (2016).
25. I. Vázquez-García *et al.*, Ovarian cancer mutational processes drive site-specific immune evasion. *Nature* **612**, 778-786 (2022).
26. J. Qian *et al.*, A pan-cancer blueprint of the heterogeneous tumor microenvironment revealed by single-cell profiling. *Cell Research* **30**, 745-762 (2020).
27. M. Büttner, J. Ostner, C. L. Müller, F. J. Theis, B. Schubert, scCODA is a Bayesian model for compositional single-cell data analysis. *Nature Communications* **12**, 6876 (2021).
28. L. Heumos *et al.*, Best practices for single-cell analysis across modalities. *Nature Reviews Genetics* **24**, 550-572 (2023).
29. Y. Cao *et al.*, scDC: single cell differential composition analysis. *BMC Bioinformatics* **20**, 721 (2019).
30. C. Cui *et al.*, Ratio of the interferon- γ signature to the immunosuppression signature predicts anti-PD-1 therapy response in melanoma. *npj Genomic Medicine* **6**, 7 (2021).
31. I. Tirosh *et al.*, Dissecting the multicellular ecosystem of metastatic melanoma by single-cell RNA-seq. *Science* **352**, 189-196 (2016).
32. W. M. Lambers *et al.*, Interferon score is increased in incomplete systemic lupus erythematosus and correlates with myxovirus-resistance protein A in blood and skin. *Arthritis Research & Therapy* **21**, 260 (2019).
33. G. Hou *et al.*, SLE non-coding genetic risk variant determines the epigenetic dysfunction of an immune cell specific enhancer that controls disease-critical microRNA expression. *Nature Communications* **12**, 135 (2021).
34. G. Finak *et al.*, MAST: a flexible statistical framework for assessing transcriptional changes and characterizing heterogeneity in single-cell RNA sequencing data. *Genome Biology* **16**, 278 (2015).

35. A. Subramanian *et al.*, Gene set enrichment analysis: A knowledge-based approach for interpreting genome-wide expression profiles. *Proceedings of the National Academy of Sciences* **102**, 15545-15550 (2005).
36. V. K. Mootha *et al.*, PGC-1 α -responsive genes involved in oxidative phosphorylation are coordinately downregulated in human diabetes. *Nature Genetics* **34**, 267-273 (2003).
37. A. Liberzon *et al.*, The Molecular Signatures Database (MSigDB) hallmark gene set collection. *Cell Syst* **1**, 417-425 (2015).
38. S. Jin *et al.*, Inference and analysis of cell-cell communication using CellChat. *Nature Communications* **12**, 1088 (2021).
39. N. Niknafs *et al.*, Persistent mutation burden drives sustained anti-tumor immune responses. *Nature Medicine* **29**, 440-449 (2023).
40. Y.-J. Kang *et al.*, A scoping review and meta-analysis on the prevalence of pan-tumour biomarkers (dMMR, MSI, high TMB) in different solid tumours. *Scientific Reports* **12**, 20495 (2022).
41. H. Rizvi *et al.*, Molecular Determinants of Response to Anti-Programmed Cell Death (PD)-1 and Anti-Programmed Death-Ligand 1 (PD-L1) Blockade in Patients With Non-Small-Cell Lung Cancer Profiled With Targeted Next-Generation Sequencing. *J Clin Oncol* **36**, 633-641 (2018).
42. S. Matsumura, S. Demaria, Up-regulation of the pro-inflammatory chemokine CXCL16 is a common response of tumor cells to ionizing radiation. *Radiat Res* **173**, 418-425 (2010).
43. D. H. Peng *et al.*, Collagen promotes anti-PD-1/PD-L1 resistance in cancer through LAIR1-dependent CD8⁺ T cell exhaustion. *Nature Communications* **11**, 4520 (2020).
44. L. C. Platanius, Mechanisms of type-I- and type-II-interferon-mediated signalling. *Nature Reviews Immunology* **5**, 375-386 (2005).
45. C. Zhou, Y. Gao, P. Ding, T. Wu, G. Ji, The role of CXCL family members in different diseases. *Cell Death Discovery* **9**, 212 (2023).
46. A. Claude-Taupin *et al.*, ATG9A protects the plasma membrane from programmed and incidental permeabilization. *Nature Cell Biology* **23**, 846-858 (2021).
47. C.-Y. Jiang *et al.*, Deregulation of ATG9A by impaired AR signaling induces autophagy in prostate stromal fibroblasts and promotes BPH progression. *Cell Death & Disease* **9**, 431 (2018).
48. A. Claude-Taupin *et al.*, ATG9A Is Overexpressed in Triple Negative Breast Cancer and Its In Vitro Extinction Leads to the Inhibition of Pro-Cancer Phenotypes. *Cells* **7**, (2018).
49. Q. Dan, Y. Yang, H. Ge, cGAS-STING Pathway as the Target of Immunotherapy for Lung Cancer. *Curr Cancer Drug Targets* **23**, 354-362 (2023).
50. H. Wang *et al.*, cGAS is essential for the antitumor effect of immune checkpoint blockade. *Proc Natl Acad Sci U S A* **114**, 1637-1642 (2017).
51. L. Lu *et al.*, STING signaling promotes NK cell antitumor immunity and maintains a reservoir of TCF-1⁺ NK cells. *Cell Reports* **42**, 113108 (2023).
52. T. Saitoh *et al.*, Atg9a controls dsDNA-driven dynamic translocation of STING and the innate immune response. *Proc Natl Acad Sci U S A* **106**, 20842-20846 (2009).
53. D. W. Vredevoogd, G. Apriamashvili, D. S. Peeper, The (re)discovery of tumor-intrinsic determinants of immune sensitivity by functional genetic screens. *Immuno-Oncology and Technology* **11**, 100043 (2021).
54. S. Li *et al.*, STING-induced regulatory B cells compromise NK function in cancer immunity. *Nature* **610**, 373-380 (2022).
55. A. Hu *et al.*, Harnessing innate immune pathways for therapeutic advancement in cancer. *Signal Transduction and Targeted Therapy* **9**, 68 (2024).

Supplementary Files

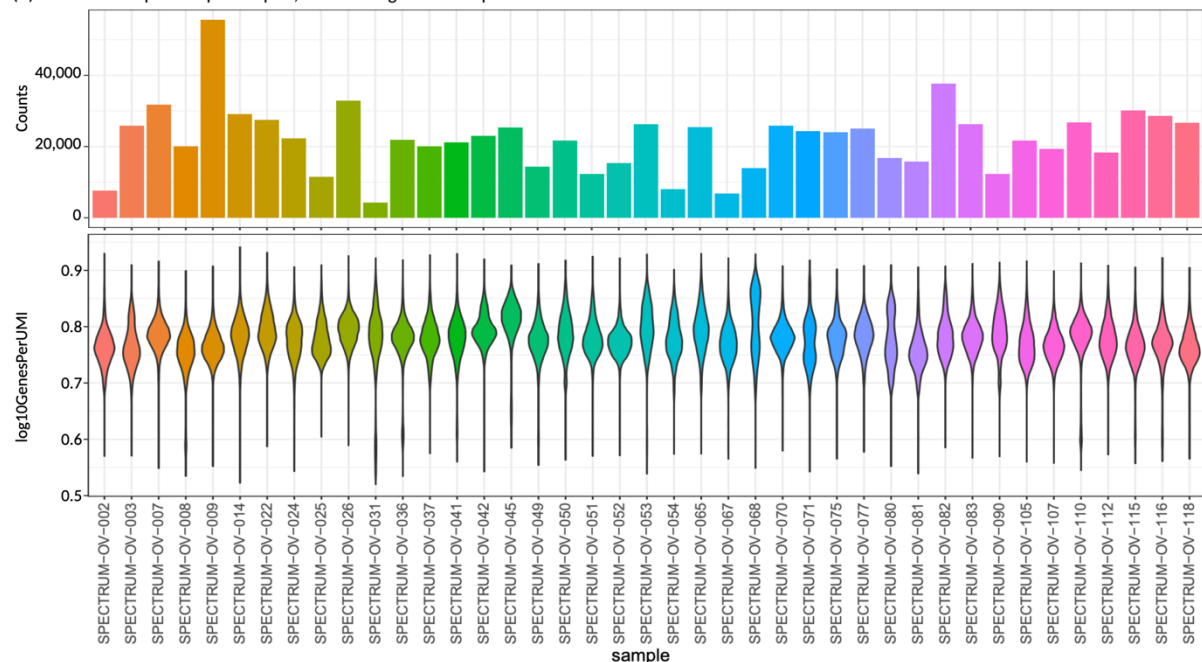
Supplementary Figure 1



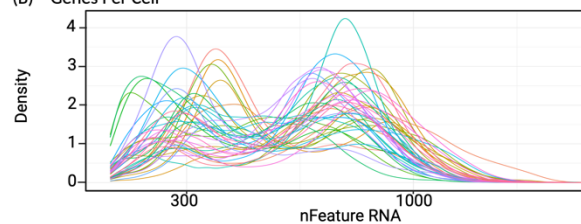
Supplementary Figure 1 | Patient Characteristics of All 6 Datasets with Bulk RNA sequencing processing (A) Shows the age distribution across six datasets with a red line marking the median age. (B) Displays TMB distribution; note that TMB for hh_lung is missing, and TMB values are log-transformed for clarity. (C) Illustrates the stage distribution, highlighting that detailed stage information is unavailable for hh_lung and mskcc_lung, but all patients are in advanced NSCLC stages. (D-E) Present HS-metrics for hh_lung and hh_ova, including bait and target coverage comparison, GC vs. AT dropout rates, coverage threshold percentages (higher percentages above 50x are preferable), and median target coverage histogram. (F) Density plot showing sample purity and ploidy estimates from FACETS. (G) Example of FACETS estimated copy number plot (H) qplot examples of amplifications and deletions from GISTIC2.

Supplementary Figure 2

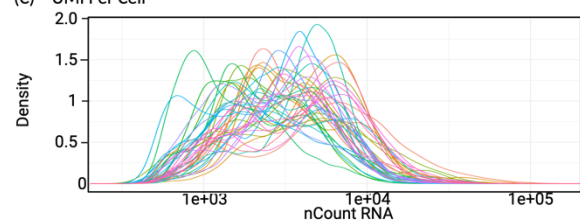
(A) Cell Counts per Sample bar plot; n cells vs n genes violin plot



(B) Genes Per Cell

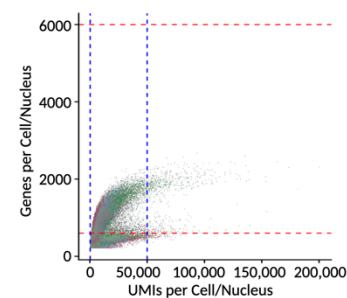


(C) UMI Per Cell

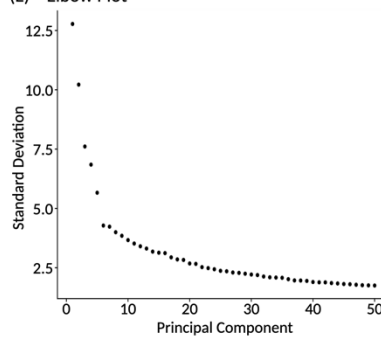


(D) Genes vs. UMIs per Cell/Nucleus

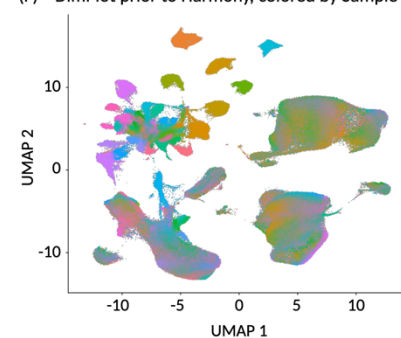
Correlation of Full Dataset is 0.69
Correlation of Filtered Dataset would be: 0.78



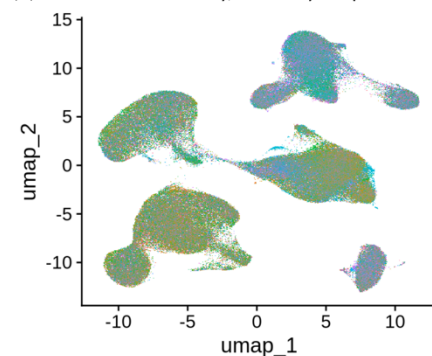
(E) Elbow Plot



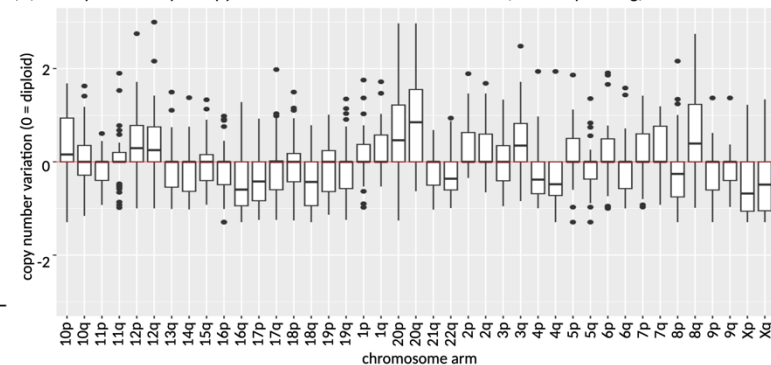
(F) DimPlot prior to Harmony, colored by Sample



(G) DimPlot After Harmony, colored by Sample



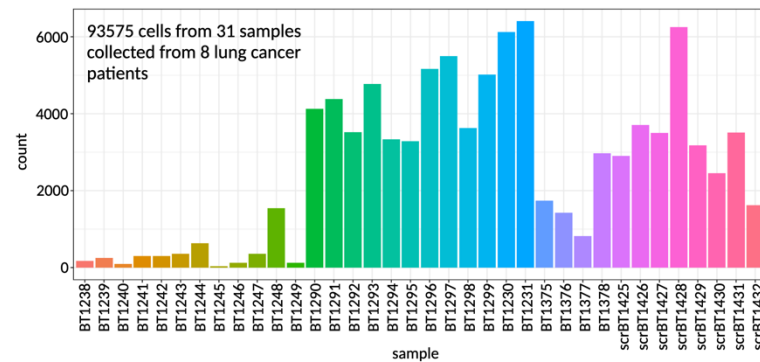
(H) Box plot of sample copy number for each chromosome arm (WGS sequencing)



Supplementary Figure 2 | Multi-faceted Quality Control and Analysis Metrics of Ovarian Cancer Single-Cell Sequencing Data. (A) Bar plot and violin plot showing cell counts per sample and the distribution of detected genes per cell, respectively. (B) Density plot illustrating the distribution of unique molecular identifiers (UMIs) per cell. (C) Density plot illustrating the distribution of unique molecular identifiers (UMIs) per cell. (D) Scatter plot comparing the correlation of genes versus UMIs per cell/nucleus before and after data filtering; post-filtering correlation is 0.78. Calculated by scCustomize package. (E) Elbow plot identifying the optimal number of principal components for downstream analysis. (F) UMAP plot before Harmony normalization, colored by sample. (G) UMAP plot after Harmony normalization, demonstrating batch effect correction, colored by sample. (H) Box plot detailing the sample copy number variation for each chromosome arm, derived from GISTIC2 algorithm using WGS copy number segmentation file downloaded from cBioPortal.

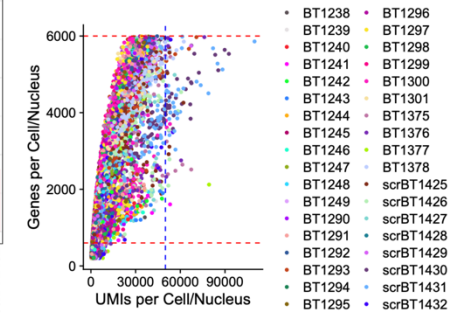
Supplementary Figure 3

(A) Cell Count per Sample

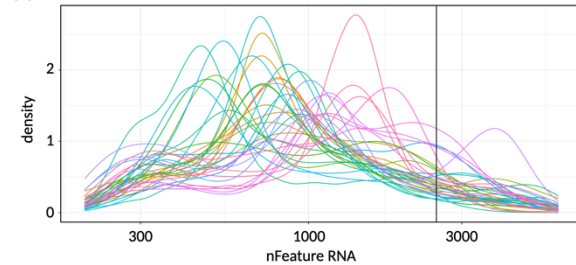


(B) Gene vs. UMI per Cell/Nucleus

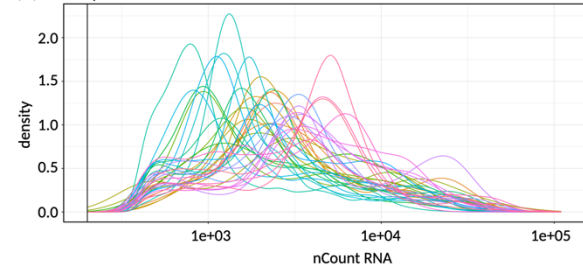
Correlation of full dataset is 0.89
Correlation of filtered dataset would be: 0.9



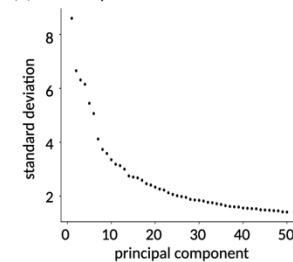
(C) Gene Per Cells



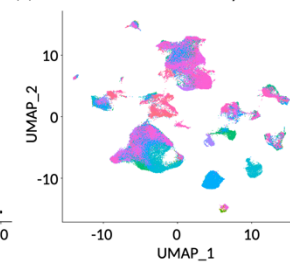
(D) UMI per cells



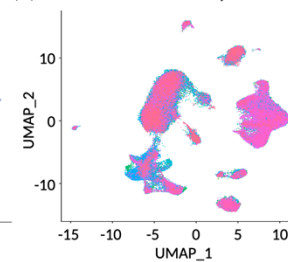
(E) Elbow plot



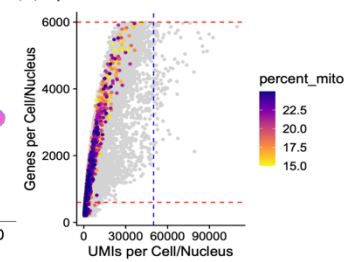
(F) DimPlot before Harmony



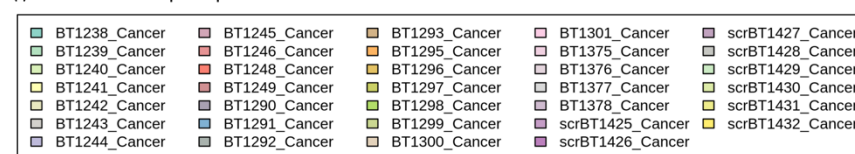
(G) DimPlot after Harmony



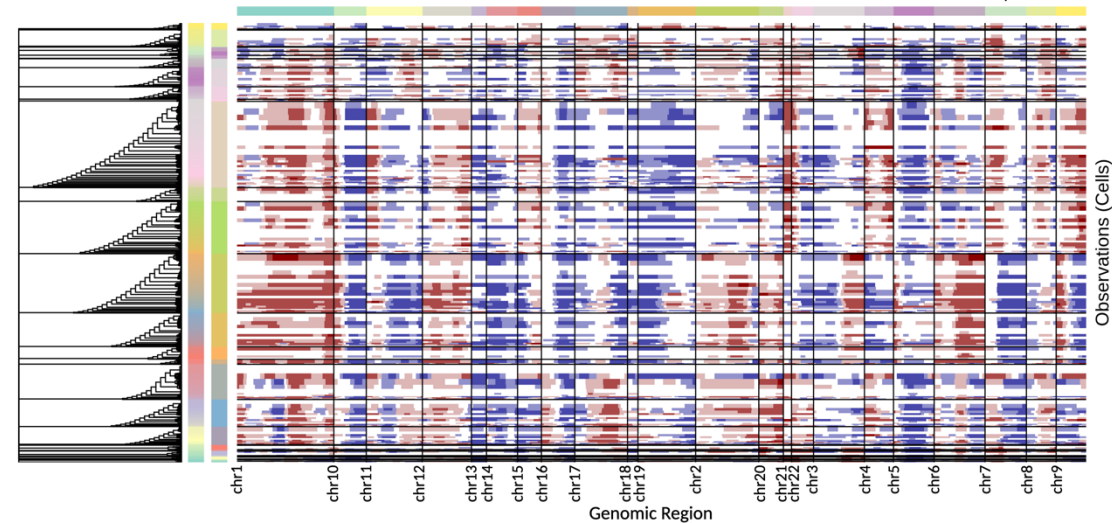
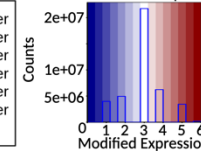
(H) percent mito



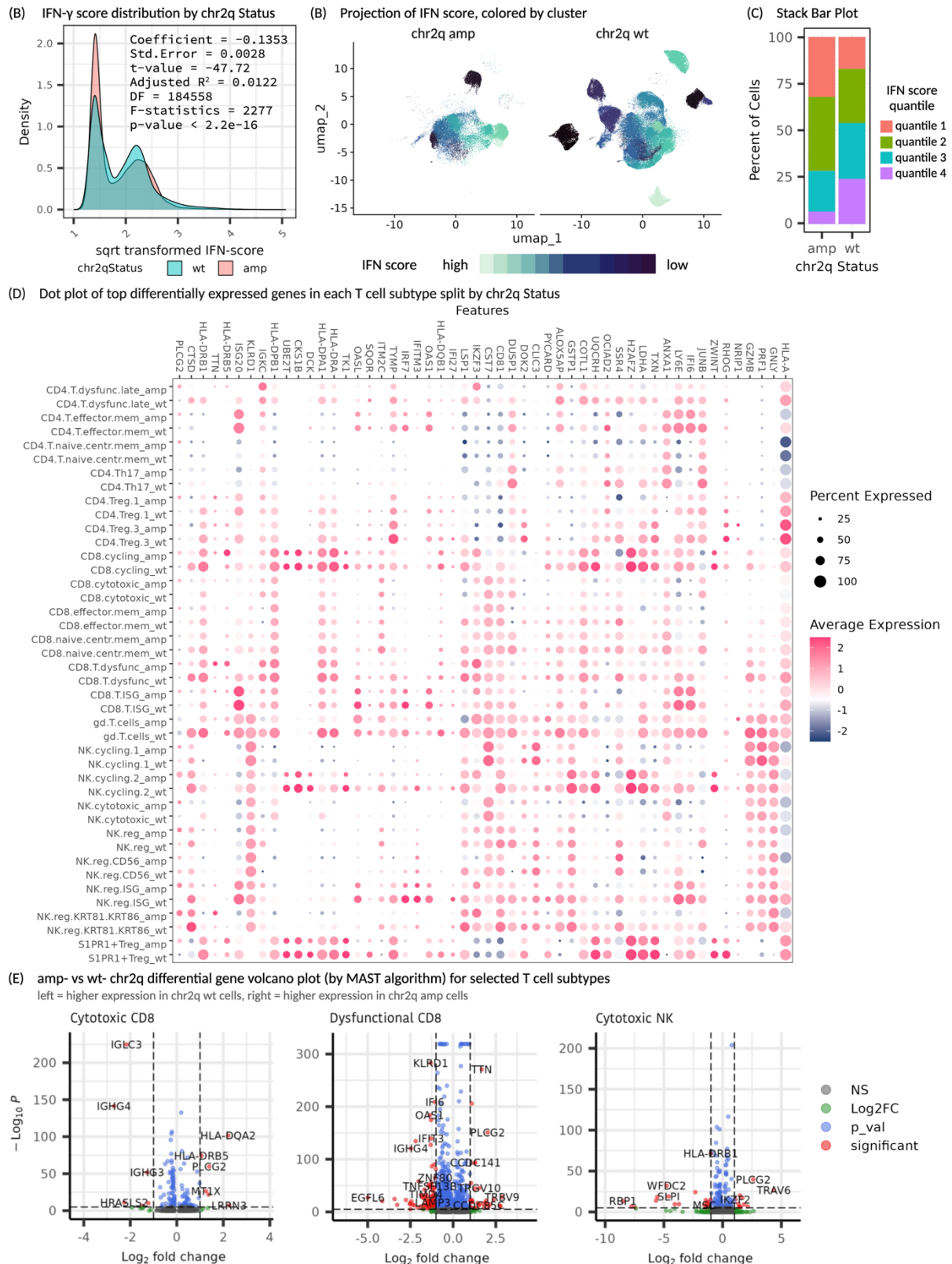
(I) Infer CNV HMM preds plot



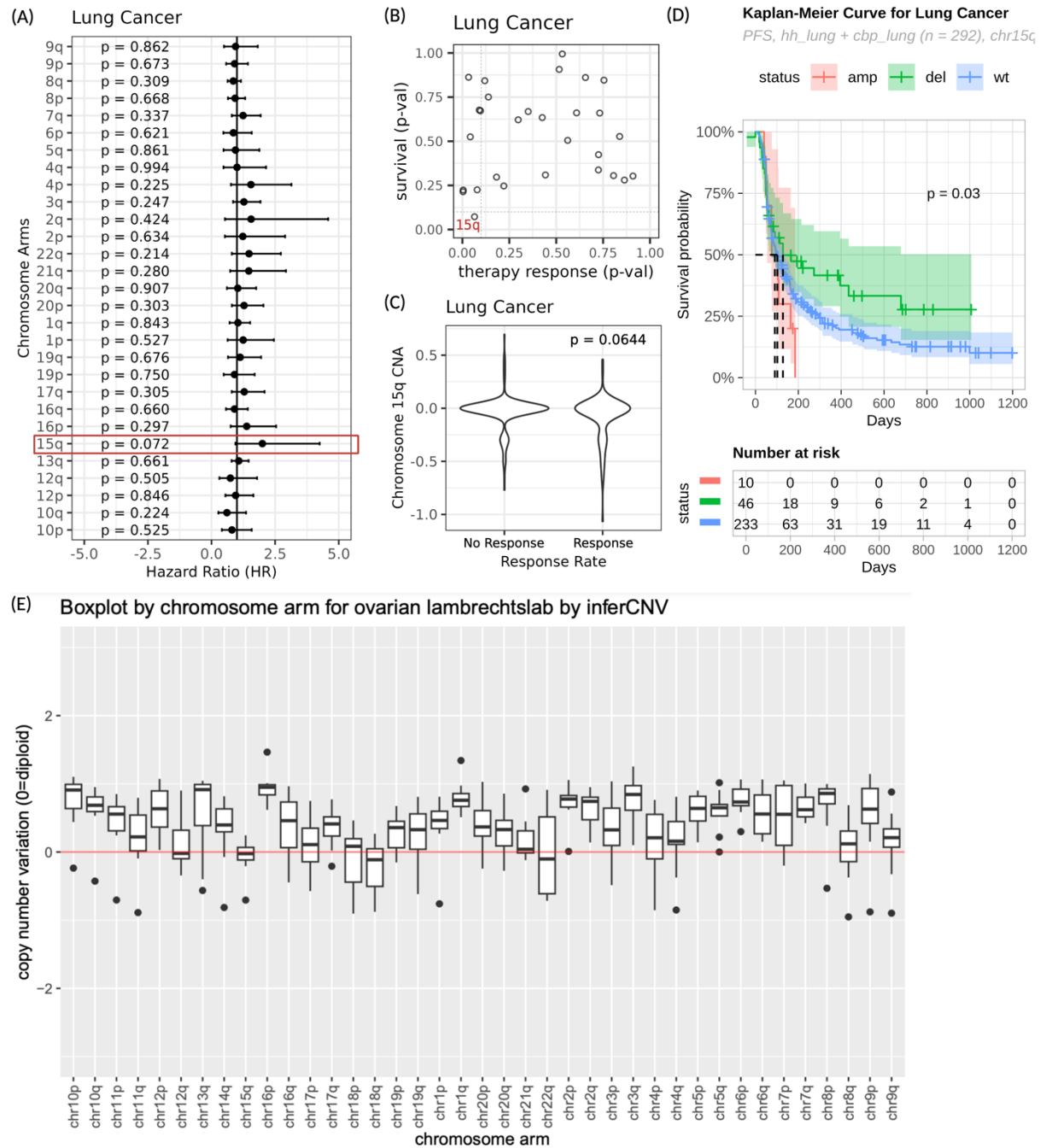
Distribution of Expression



Supplementary Figure 3 | Single-Cell RNA Sequencing Data Analysis for lung cancer (A) Bar graph depicting the number of cells per sample across different datasets. (B) Scatter plot illustrating the relationship between gene expression and UMI count per cell/nucleus, with color coding for each sample and a correlation coefficient provided for full and filtered datasets. (C) Density plot showing the distribution of genes per cell. (D) Density plot of UMI count per cell. (E) Elbow plot determining the number of principal components. (F) UMAP visualization before batch correction with Harmony. (G) UMAP visualization after batch correction with Harmony. (H) Scatter plot of percent mitochondrial gene content versus UMI count per cell/nucleus, color-coded by mitochondrial gene content. (I) Hidden Markov Model (HMM) predictions plot for inferred CNV events across genomic regions, with each row representing a cell and color intensity indicating the modified expression level.

Supplementary Figure 4

Supplementary Figure 4 | Characterization of IFN and IFN- γ Score Profiles and T cell differentially expressed genes. (A) A scatter plot delineates the IFN- γ score stratified by chromosome 2q status, with linear regression $\text{lm}(\text{formula} = \text{sqr_IFN_score} \sim \text{chr2q}, \text{data} = \text{IFN_score})$ revealing a statistically significant divergence between amplified and wildtype groups. Notably, a negative correlation was discerned (regression coefficient = -0.1353), indicating a decrease in normalized IFN- γ scores concomitant with elevated chr2q copy numbers. (B) UMAP representation highlights the distribution of IFN scores, with intensity correlating to higher score values. (C) A stacked bar chart quantifies the cellular composition within IFN score quartiles. (D) Dot plot illustrating gene expression differences within T cell subpopulations by chromosome 2q status (amplified, chr2q > 0; wildtype, chr2q $\leq$ 0). Dot size indicates the proportion of cells expressing a gene; color intensity reflects average expression, with pink for upregulation and blue for downregulation. (E) Volcano plot displaying differentially expressed genes in cytotoxic CD8, dysfunctional CD8, and cytotoxic NK cells. The significance threshold ($p < 0.05$) is shown by a dashed horizontal line. Dot colors signify: grey for non-significant changes, green for changes with a log₂ fold change (FC) ≥ 1 but not significant, blue for significant changes with log₂FC < 1, and red for significant changes with log₂FC ≥ 1 .

Supplementary Figure 5

Supplementary Figure 5 | Bulk RNA and single cell RNA sequencing of Lung Cancer. (A) Forest plot patient survival hazard ratio when arm-level CNA increases, modeled by proportional hazards regression model (coxph) and hazard ratio taken along with upper and lower 95% confidence interval, false discovery rate calculated. (B) Scatter plot of p-value for difference in arm level CNA by survival (coxph) and therapy response rate (by wilcox test). (C) Violin plot for a zoomed in view of the arm level CNA by therapy response (response vs no response) in selected chromosome arm. (D) Kaplan-Meier survival plot of ovarian and lung cancer patient with chromosome 2q and 15q CNA respectively. Plot created by ggsurvplot and p-value computed by log-rank test using survdiff with weights == 1. (E) Paneled box plot of chromosome arm copy number distribution with copy number inferred from single cell sequencing data usi

Supplementary Table 1

(A) Differential Cell Composition analysis data for Myeloid cells and T cell produced by scCODA analysis

cell.type	term	Final Parameter	HDI 3%	HDI 97%	SD	Inclusion probability	Expected Sample	log2- FC
T cells	CD4.T.CXCL13	0.0	-0.163	0.279	0.080	0.4906	197.981	0.0
	CD4.T.naive	0.0	-0.161	0.114	0.047	0.4277	1069.738	0.0
	CD4.T.reg	0.0	-0.146	0.172	0.055	0.4493	585.326	0.0
	CD8.T.CXCL13	0.0	-0.206	0.186	0.069	0.4642	293.586	0.0
	CD8.T.ISG	0.0	-0.168	0.179	0.060	0.4411	440.174	0.0
	CD8.T.cytotoxic	0.0	-0.207	0.077	0.059	0.4767	1257.863	0.0
	Cycling.T.NK	0.0	0.000	0.000	0.000	0.0000	369.138	0.0
	ILC	0.0	-0.213	0.217	0.077	0.4702	153.418	0.0
	NK.CD56	0.0	-0.098	0.241	0.071	0.4868	658.635	0.0
	NK.cytotoxic	0.0	-0.328	0.102	0.098	0.5383	463.668	0.0

(B) Differential Cell Composition analysis data for Myeloid cells and T cell produced by scDC analysis

cell.type	term	estimate	std.error	statistic	df	p.value	FDR
Myeloid cells	condwt	-0.056	0.020	-2.730	9.516	2.2E-02	8.9E-02
	subjectwt	0.450	0.020	22.416	9.812	9.4E-10	9.4E-09
	cellTypesM1:condwt	0.265	0.042	6.292	9.891	9.4E-05	6.6E-04
	cellTypesM10:condwt	-1.372	0.413	-3.321	7.636	1.1E-02	5.6E-02
	cellTypesM2:condwt	0.064	0.044	1.471	9.467	1.7E-01	5.2E-01
	cellTypesM3:condwt	-0.252	0.027	-9.485	9.330	4.3E-06	3.9E-05
	cellTypesM4:condwt	-0.049	0.048	-1.013	8.352	3.4E-01	6.5E-01
	cellTypesM5:condwt	-1.048	0.038	-27.428	9.748	1.5E-10	2.1E-09
	cellTypesM6:condwt	-0.209	0.052	-4.009	9.042	3.0E-03	1.8E-02
	cellTypesM7:condwt	-0.039	0.038	-1.037	9.842	3.2E-01	6.5E-01
T cells	cellTypesM8:condwt	NA	NA	NA	NA	NA	NA
	cellTypesM9:condwt	-0.339	0.045	-7.605	9.130	3.1E-05	2.4E-04
	condwt	0.105	0.029	3.661	19.961	1.6E-03	2.3E-02
	subjectwt	0.827	0.030	27.598	18.005	3.5E-16	1.3E-14
	cellTypesCD4.T.effector.mem:condwt	0.673	0.065	10.351	20.249	1.6E-09	4.0E-08
	cellTypesCD4.T.naive.cent.mem:condwt	0.036	0.049	0.744	16.550	4.7E-01	1.0E+00
	cellTypesCD4.Th17:condwt	-0.098	0.052	-1.872	20.943	7.5E-02	6.0E-01
	cellTypesCD4.Treg.1:condwt	-0.093	0.049	-1.879	16.895	7.8E-02	6.0E-01
	cellTypesCD4.Treg.3:condwt	0.144	0.065	2.226	15.949	4.1E-02	4.1E-01
	cellTypesCD8.cytotoxic:condwt	0.143	0.045	3.181	18.559	5.0E-03	6.6E-02
	cellTypesCD8.effector.mem:condwt	-0.289	0.049	-5.846	19.503	1.1E-05	2.1E-04
	cellTypesCD8.naive.cent.mem:condwt	-0.522	0.055	-9.434	15.797	6.9E-08	1.6E-06
	cellTypesCD8.T.dysfunc:condwt	-0.346	0.048	-7.276	19.098	6.5E-07	1.4E-05
	cellTypesCD8.T.ISG:condwt	-0.125	0.050	-2.481	19.335	2.2E-02	2.7E-01
	cellTypescycling.CD8:condwt	-0.113	0.057	-2.007	19.139	5.9E-02	5.3E-01
	cellTypescycling.NK.1:condwt	0.504	0.049	10.208	17.663	7.8E-09	1.9E-07
	cellTypescycling.NK.2:condwt	-0.033	0.080	-0.418	17.211	6.8E-01	1.0E+00
	cellTypessgd.T.cells:condwt	-1.952	0.274	-7.119	16.726	1.9E-06	3.8E-05
	cellTypesNK.cytotoxic:condwt	-0.021	0.057	-0.364	19.549	7.2E-01	1.0E+00
	cellTypesNK.reg:condwt	-0.266	0.052	-5.116	16.628	9.2E-05	1.7E-03
	cellTypesNK.reg.CD56:condwt	-0.349	0.067	-5.193	14.706	1.2E-04	2.0E-03
	cellTypesNK.reg.ISG:condwt	-0.763	0.061	-12.410	18.807	1.7E-10	4.7E-09
	cellTypesNK.reg.KRT81.KRT86:condwt	NA	NA	NA	NA	NA	NA
	cellTypesS1PR1+Treg:condwt	-0.144	0.103	-1.398	16.811	1.8E-01	8.9E-01

Supplementary Table 1 | (A) Differential Cell Composition analysis with scCODA framework for T cells sub cell types, the table demonstrates the basic statistics calculated by scCODA framework. Reported statistics includes final parameter, high density interval (HDI), standard deviation (SD), inclusion probability, expected samples, and log2 fold change (log2FC). Upon further credible effect analysis with FDR set to 0.20, no significant cell types were identified. (B) Differential cell composition analysis with scDC framework for myeloid and T cell sub cell types, the table demonstrates the statistics visualised by applying general linear model on scDC object and the resulting estimates, standard error (std.error), statistics, degree freedom (df), p-value and false-discovery rate values.

Supplementary Table 2

Communication	Chromosome 2q Status	CellChat Discovered Pathways
Cancer to Myeloid	Amplified	MHC-II, ANNEXIN, ApoE, CCL, COMPLEMENT, GALECTIN, LAIR1, PECAM1, SPP1, MK, ADGRE, GRN, GAS, PLAUI, ApoA, FN1, THBS, CD39, PECAM2, CLDN, ICAM, SN, CADM, EGF, SELPLG, CHEMERIN, LAMININ, PTN, SELL, COLLAGEN, AGRN, SAA, ADGRG, EPHB, TNF, NECTIN, CDH, RESISTIN, DESMOSOME, SEMA4, IGF, HSPG, FGF, JAM, PARS
	Wild Type	MHC-II, ANNEXIN, ApoE, CCL, COMPLEMENT, GALECTIN, LAIR1, MK, PLAUI, GRN, GAS, FN1, PECAM2, CLDN, PECAM1, CD39, SPP1, ADGRE, CADM, RESISTIN, ICAM, ApoA, THBS, SN, NECTIN, ADGRG, SELL, CHEMERIN, TNF, EPHB, SELPLG, COLLAGEN, LAMININ, PTN, DESMOSOME, AGRN, SEMA4, CDH, IGF, HSPG, JAM, NOTCH
Myeloid to T Cells	Amplified	MHC-I, MHC-II, CLEC, CCL, ANNEXIN, ApoE, IFN-II, CD6, LCK, ICAM, SPP1, ADGRE, FN1, CXCL, PECAM2, TNF, CD80, FASLG, CD40, COMPLEMENT, CD86, LAIR1, PECAM1, PARS, SN, NECTIN, GRN, SELPLG, CSF, MK, CD39, THBS, CADM, IL1, RESISTIN, NCAM, GALECTIN, ADGRG, CHEMERIN, CLDN, SAA, COLLAGEN, JAM
	Wild Type	MHC-I, MHC-II, CCL, CLEC, ANNEXIN, LCK, ApoE, GALECTIN, CXCL, ADGRE, CD86, COMPLEMENT, FN1, LAIR1, PECAM2, SELPLG, SPP1, PLAUI, CD6, PARS, CD39, CADM, FASLG, TNF, CD70, PECAM1, GRN, RESISTIN, IFN-II, CSF, IL1, SN, MK, THBS, ICAM, NECTIN, GAS, LT, CD40, ADGRG, CLDN, NCAM, COLLAGEN, CDH1, DESMOSOME, CHEMERIN, LAMININ, AGRN, CDH, KLK, IGF, JAM, NOTCH

Supplementary Table 2 | Cell chat discovered pathways between cancer cells to myeloid cells and myeloid cells to T cells, differentiated by chromosome 2q status.

Supplementary Method***Hammersmith Dataset Sequencing Metrics***

The customized hg38.fasta reference was built by selecting the following chromosomal and genomic sequences in GoldenPath GRCh38 directory. Chromosome 1 to 22, MT, X, Y, KI270733.1, KI270726.1, KI270467.1, KI270387.1, GL000224.1, KI270735.1, KI270362.1, KI270391.1; GL000219.1, KI270711.1, KI270517.1, KI270305.1; KI270719.1, KI270745.1, KI270593.1, KI270373.1, GL000216.2, KI270714.1, KI270528.1, KI270422.1, KI270712.1, KI270732.1, KI270587.1, KI270316.1; KI270706.1, KI270713.1, KI270364.1, KI270340.1; KI270725.1, KI270754.1, KI270371.1, KI270338.1, KI270744.1, KI270710.1, KI270333.1, KI270583.1, KI270734.1, KI270717.1, KI270374.1, KI270334.1, GL000213.1, KI270724.1, KI270411.1, KI270429.1, GL000220.1, KI270720.1, KI270414.1, KI270393.1; KI270715.1, KI270723.1, KI270510.1, KI270516.1; GL000218.1, KI270718.1, KI270390.1, KI270389.1; KI270749.1, KI270317.1, KI270375.1, KI270466.1; KI270741.1, KI270740.1, KI270420.1, KI270388.1, GL000221.1, KI270755.1, KI270509.1, KI270544.1, KI270716.1, KI270707.1, KI270315.1, KI270310.1; KI270731.1, KI270579.1, KI270302.1, KI270412.1; 7 KI270751.1, KI270752.1, KI270518.1, KI270395.1; KI270750.1, KI270512.1, KI270530.1, KI270376.1, KI270519.1, KI270322.1, KI270304.1, KI270337.1, GL000214.1, GL000226.1, KI270418.1, KI270335.1, KI270708.1, KI270311.1, KI270424.1, KI270378.1, KI270730.1, KI270366.1, KI270417.1, KI270379.1; KI270728.1, KI270438.1, KI270511.1, KI270508.1, KI270329.1, KI270727.1, KI270737.1, KI270448.1, KI270303.1, KI270419.1; KI270442.1, KI270721.1, KI270521.1, KI270381.1, KI270336.1, KI270729.1, KI270738.1, KI270581.1, KI270529.1, KI270312.1; GL000225.1, KI270748.1, KI270582.1, KI270425.1, KI270539.1, KI270743.1, KI270435.1, KI270515.1, KI270396.1, KI270385.1; GL000008.2, GL000208.1, KI270588.1, KI270363.1, KI270423.1; GL000009.2, KI270538.1, KI270591.1, KI270386.1, KI270392.1; KI270747.1, KI270756.1, KI270522.1, KI270465.1, KI270394.1; KI270722.1, KI270739.1, KI270507.1, KI270383.1; GL000194.1, KI270757.1, KI270590.1, KI270384.1; KI270742.1, KI270709.1, KI270584.1, KI270330.1; GL000205.2, KI270746.1, KI270320.1, KI270372.1; GL000195.1, KI270753.1, KI270382.1, KI270548.1; KI270736.1, KI270589.1, KI270468.1, KI270580.1.